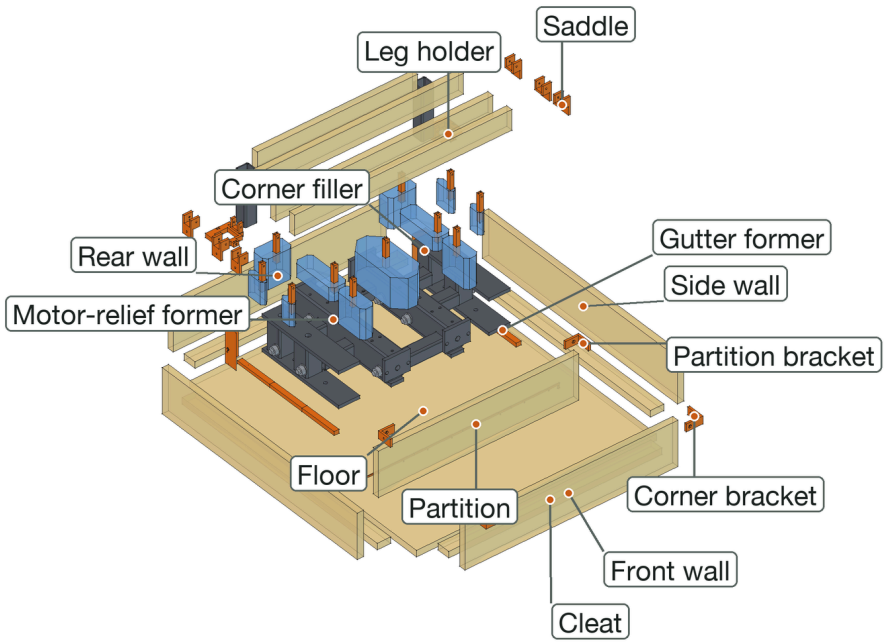
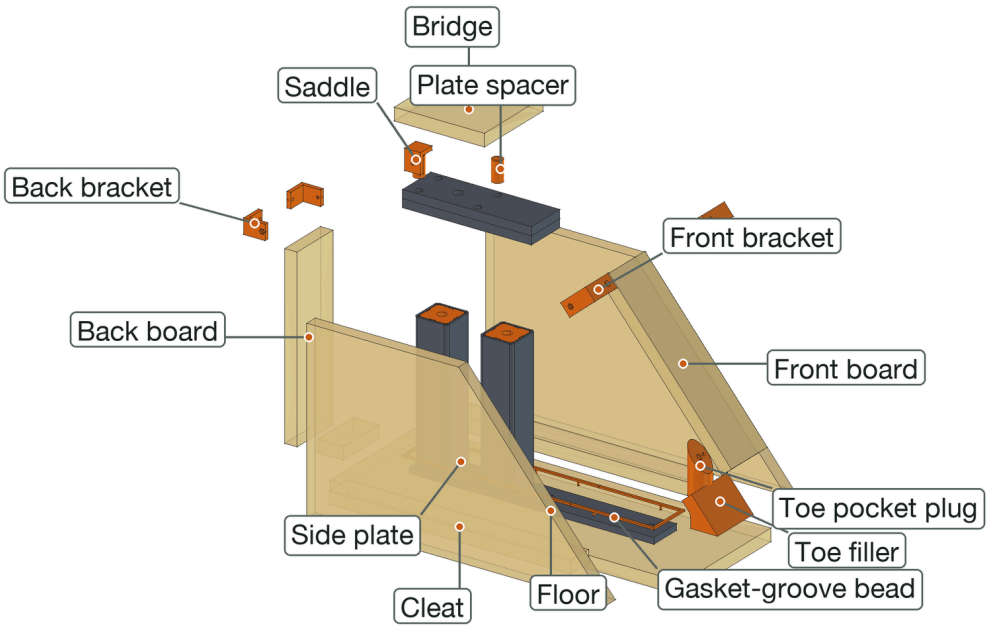
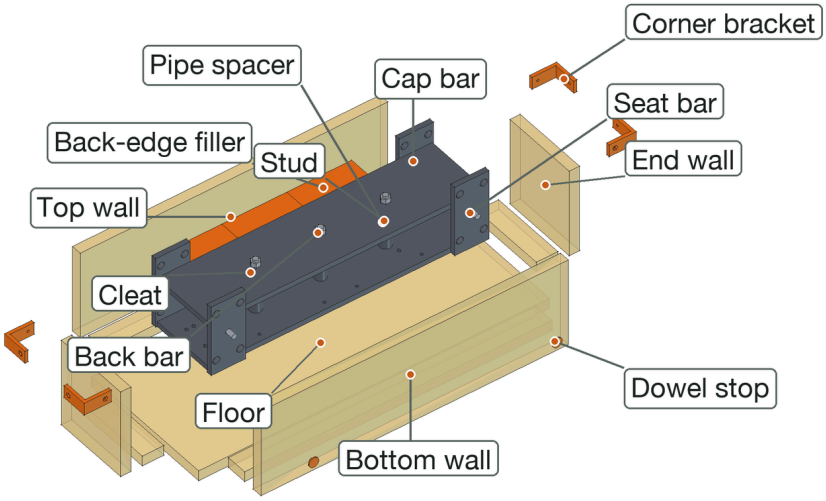
Mold guide

DIY CNC epoxy-granite mill · October 2026

How to make the molds for the five castings. Each mold also has its own shorter guide in this folder (base, uprights, beam), and its own printable drilling templates in `templates/`.





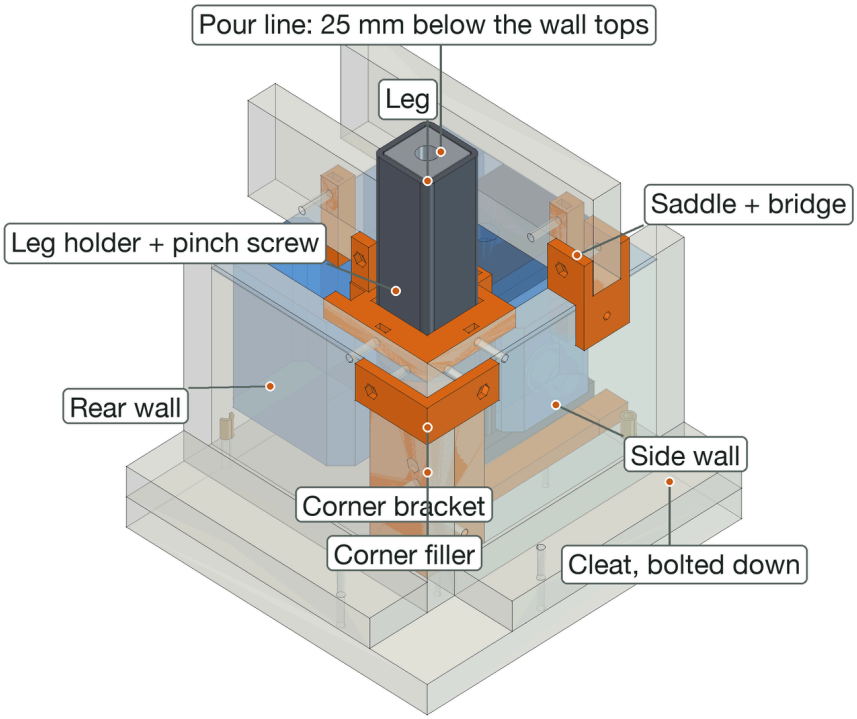


In short: melamine boards bolt together with printed brackets. Every printed piece bolts or pins to its board from outside. The walls stand 25 mm above the pour line, so no bolt goes through a face the mix touches. Nothing is glued, and it all comes apart for the next casting.

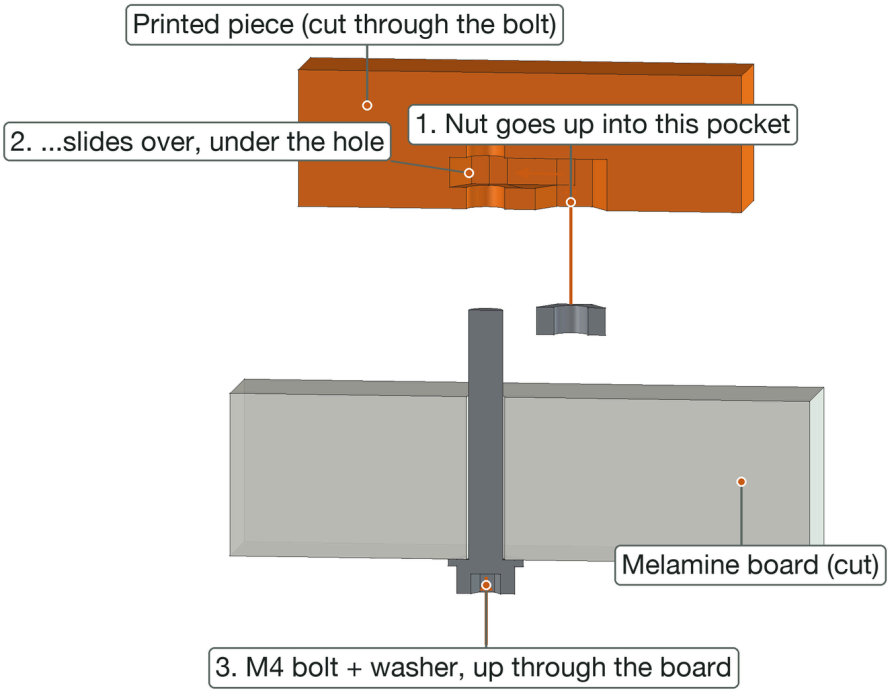
Order of work: print → cut the boards → mark them with the templates and drill → cut the foam → wax → assemble → pour → take apart.

1. How it works

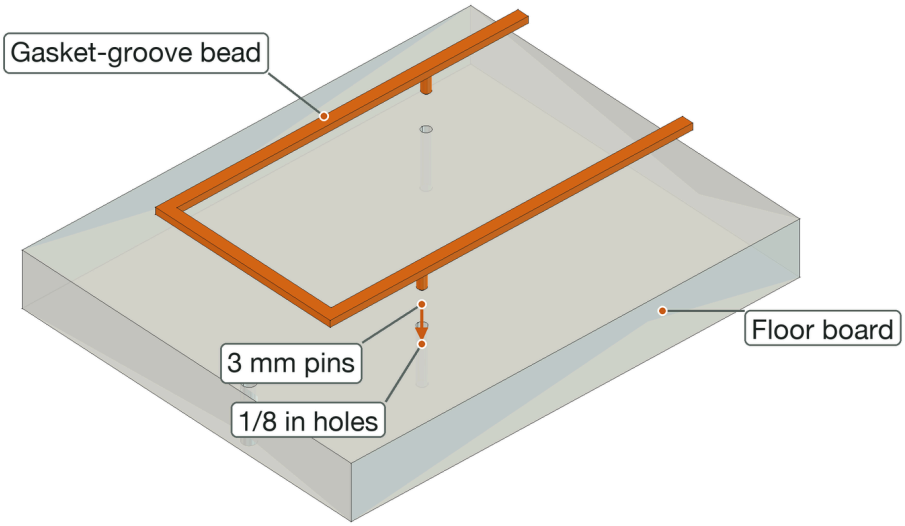
- **Walls** stand on the floor against bolted-down **cleats**; **corner brackets** square the tops. All the bolts go through the top 25 mm, above the pour line.



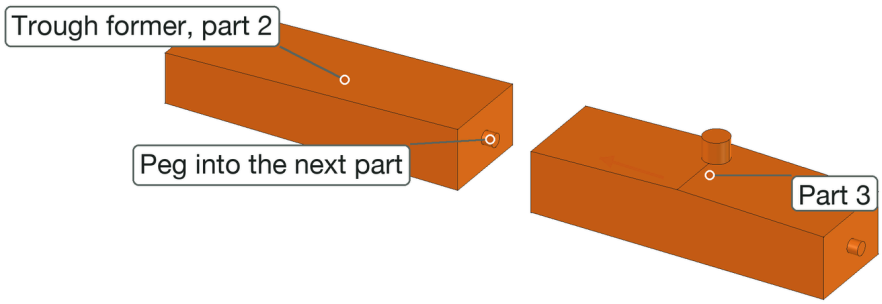
- **Printed pieces** have a nut built in: it goes into a pocket on the board side and slides under the hole; an M4 bolt comes through the board from outside.



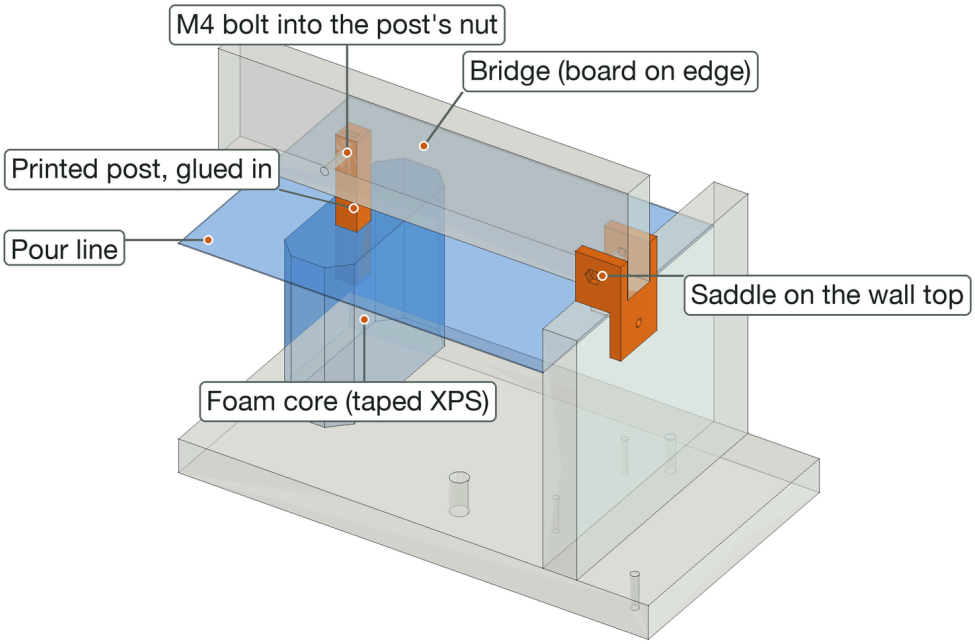
- **Thin pieces** (gasket-groove beads, motor relief) have 3 mm pins that push into 1/8 in holes.



- **Long pieces** are printed in parts that peg together; each part bolts down on its own.



- **Cores** are taped foam blocks with a printed post that bolts to a bridge across the mold.



- **Fits** are 0.15 mm a side, and drawn for 19.05 mm board (Home Depot's 3/4 in melamine is a true 0.750 in). Measure yours; if it differs, set `mold_board` in `model/build_cnc.py` and re-run it.

2. What you need

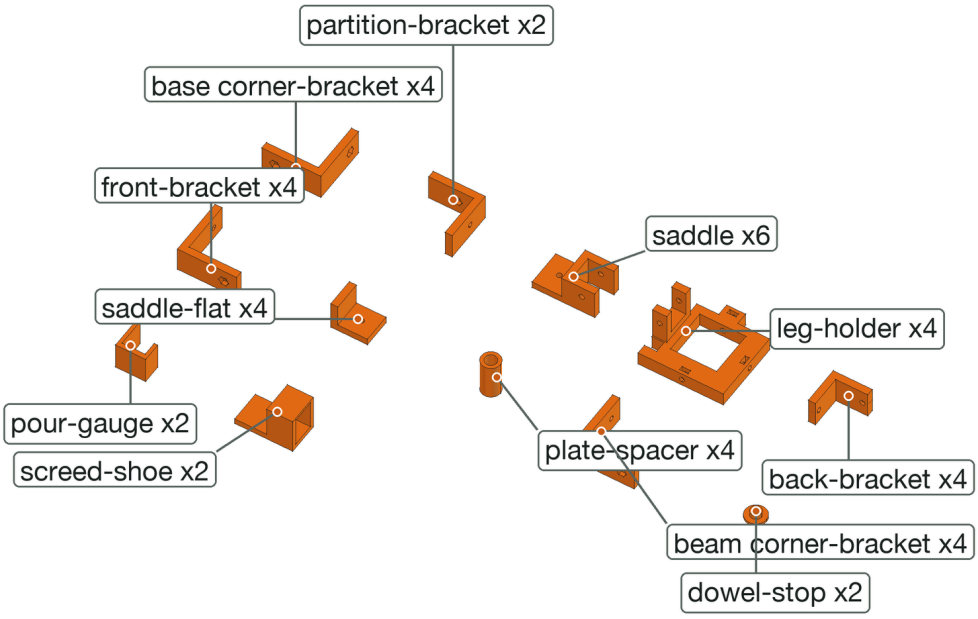
Material	Amount	Notes
Melamine board, 3/4 in	2 sheets, 4 x 8 ft	layout in section 4
XPS foam board, 2 in	1 sheet, 2 x 8 ft	the cores
PETG	1.02 kg	sliced; everything printed for the machine fits in two 1 kg rolls
Packing tape, double-sided carpet tape	1 roll each	wrapping and holding the foam
Paste wax and PVA	1 each	release
3/8 in quarter-round PVC molding	about 6 m	the rounded outside edges
2x4 offcut	4 blocks per upright mold, about 50 mm	feet

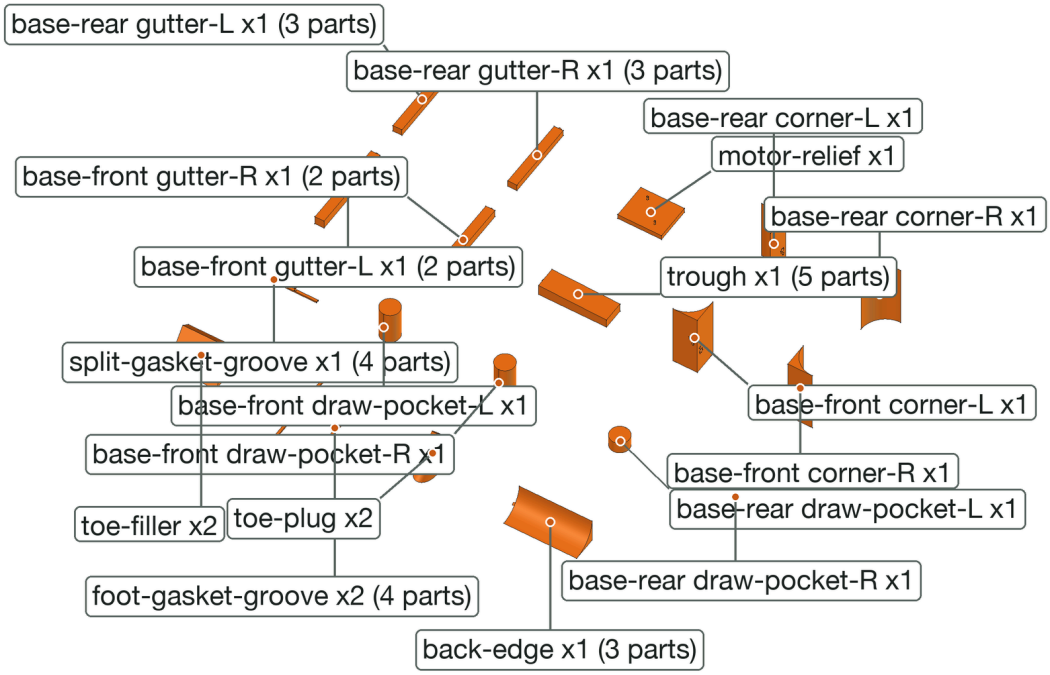
Hardware	Qty
M3 nut	20
M3 washer	20
M3 x 25 bolt	20
M4 nut	174
M4 washer	170
M4 x 20 bolt (leg pinch screw)	4
M4 x 30 bolt	98
M4 x 35 bolt	32
M4 x 45 bolt	40
M5 x 20 screw, greased (holds an insert)	11
M5 x 30 screw, greased (holds a rail bar)	10
M5 x 40 screw, greased (holds the face plate)	8
M6 x 30 screw, greased (holds a support pad)	6
M6 x 40 screw, greased (holds the face plate)	6
1/2 in plain steel rod, greased (forms a beam bolt hole), 204 mm	8
1/2-13 x 1-1/2 in bolt + nut, greased (forms a dowel hole)	4
3/8-16 x 3 in bolt, greased (holds the top plate, both layers)	4
the two column tie rods themselves, greased: up through the floor and the columns into the top plate, to center the columns while the casting goes off	2

Add about 10 % spare small bolts, nuts and washers. **Tools:** drill and brad-point bits (sizes in section 5), a 2-1/2 in hole saw, a jigsaw for the side plates' slope, a square, clamps, a 7 mm nut driver, a 3 mm hex key, an awl, a fine saw or hot wire for the foam.

3. Print

116 prints, 1.02 kg, about 84 hours (sliced; settings in the print guide). Print one corner bracket and one saddle first and check them on a board offcut. Press or slide the nuts in before the pieces go on the boards.





File	Qty	g	Time each
base-box/kit-corner-bracket.stl	4	33	39m 23s
base-box/kit-partition-bracket.stl	2	14	35m 28s
base-box/kit-saddle.stl	6	58	46m 3s
base-box/kit-leg-holder.stl	4	82	1h 37m 0s
base-rear/former-gutter-L-part1of3.stl	1	8	35m 5s
base-rear/former-gutter-L-part2of3.stl	1	8	33m 40s
base-rear/former-gutter-L-part3of3.stl	1	7	31m 15s
base-rear/former-gutter-R-part1of3.stl	1	8	35m 6s
base-rear/former-gutter-R-part2of3.stl	1	8	33m 40s
base-rear/former-gutter-R-part3of3.stl	1	7	31m 15s
base-rear/former-motor-relief.stl	1	12	40m 58s
base-rear/filler-corner-L.stl	1	17	1h 44m 24s
base-rear/filler-corner-R.stl	1	16	1h 42m 27s
base-front/former-gutter-L-part1of2.stl	1	11	49m 23s
base-front/former-gutter-L-part2of2.stl	1	10	43m 43s
base-front/former-gutter-R-part1of2.stl	1	11	49m 24s
base-front/former-gutter-R-part2of2.stl	1	10	43m 43s
base-front/former-trough-part1of5.stl	1	20	1h 14m 18s
base-front/former-trough-part2of5.stl	1	22	1h 23m 56s
base-front/former-trough-part3of5.stl	1	25	1h 33m 33s
base-front/former-trough-part4of5.stl	1	23	1h 24m 3s
base-front/former-trough-part5of5.stl	1	21	1h 17m 36s
base-front/filler-corner-L.stl	1	16	1h 43m 48s
base-front/filler-corner-R.stl	1	17	1h 40m 33s
base-rear/former-split-gasket-groove-part1of4.stl	1	2	9m 50s
base-rear/former-split-gasket-groove-part2of4.stl	1	1	8m 58s
base-rear/former-split-gasket-groove-part3of4.stl	1	1	8m 58s
base-rear/former-split-gasket-groove-part4of4.stl	1	2	9m 53s
base-front/core-center-a-post.stl	1	5	44m 56s
base-front/core-center-b-post.stl	1	5	44m 56s
base-rear/core-center-post.stl	1	5	44m 56s
base-front/core-outboard-L-post.stl	1	5	44m 53s
base-front/core-trough-band-L-post.stl	1	5	44m 54s
base-front/core-draw-rod-L-post.stl	1	5	44m 53s
base-rear/core-foot-b-L-post.stl	1	5	44m 53s

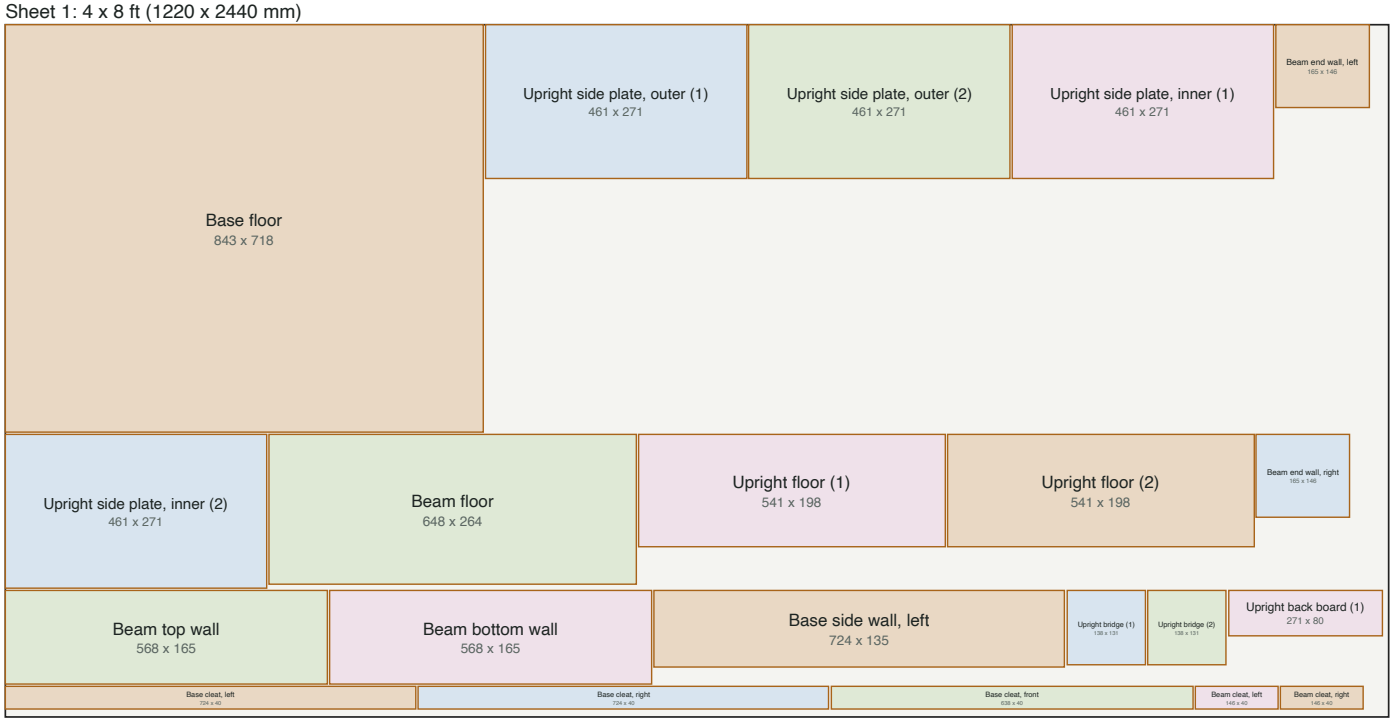
File	Qty	g	Time each
base-rear/core-knee-L-post.stl	1	5	44m 53s
base-rear/core-outer-a-L-post.stl	1	5	44m 53s
base-rear/core-outer-b-L-post.stl	1	5	44m 53s
base-rear/core-rear-L-post.stl	1	5	44m 53s
base-front/core-outboard-R-post.stl	1	5	44m 53s
base-front/core-trough-band-R-post.stl	1	5	44m 54s
base-front/core-draw-rod-R-post.stl	1	5	44m 53s
base-rear/core-foot-b-R-post.stl	1	5	44m 53s
base-rear/core-knee-R-post.stl	1	5	44m 53s
base-rear/core-outer-a-R-post.stl	1	5	44m 53s
base-rear/core-outer-b-R-post.stl	1	5	44m 53s
base-rear/core-rear-R-post.stl	1	5	44m 53s
base-front/plug-draw-pocket-L.stl	1	12	46m 16s
base-front/plug-draw-pocket-R.stl	1	12	46m 16s
base-rear/plug-draw-pocket-L.stl	1	5	21m 23s
base-rear/plug-draw-pocket-R.stl	1	5	21m 23s
upright/kit-back-bracket.stl	4	21	27m 33s
upright/kit-front-bracket.stl	4	32	38m 26s
upright/kit-saddle-flat.stl	4	18	22m 40s
upright/filler-toe-filler.stl	2	51	1h 30m 59s
upright/former-foot-gasket-groove-part1of4.stl	2	6	16m 25s
upright/former-foot-gasket-groove-part2of4.stl	2	2	8m 22s
upright/former-foot-gasket-groove-part3of4.stl	2	2	8m 22s
upright/former-foot-gasket-groove-part4of4.stl	2	6	16m 30s
upright/plug-toe-plug.stl	2	26	1h 3m 42s
upright/kit-plate-spacer.stl	4	15	20m 9s
beam/kit-corner-bracket.stl	4	33	40m 3s
beam/filler-back-edge-part1of3.stl	1	17	1h 8m 21s
beam/filler-back-edge-part2of3.stl	1	18	1h 9m 15s
beam/filler-back-edge-part3of3.stl	1	17	1h 8m 22s
beam/kit-dowel-stop.stl	2	1	4m 19s
tools/kit-pour-gauge.stl	2	9	24m 33s
tools/kit-screed-shoe.stl	2	20	44m 42s
base-front/cast-in-leg-front-l-plug.stl	2	8	24m 14s
base-rear/cast-in-leg-rear-l-plug.stl	2	8	24m 14s

File	Qty	g	Time each
`upright/cast-in-column-ring-l1a.stl`	8	109	49m 49s

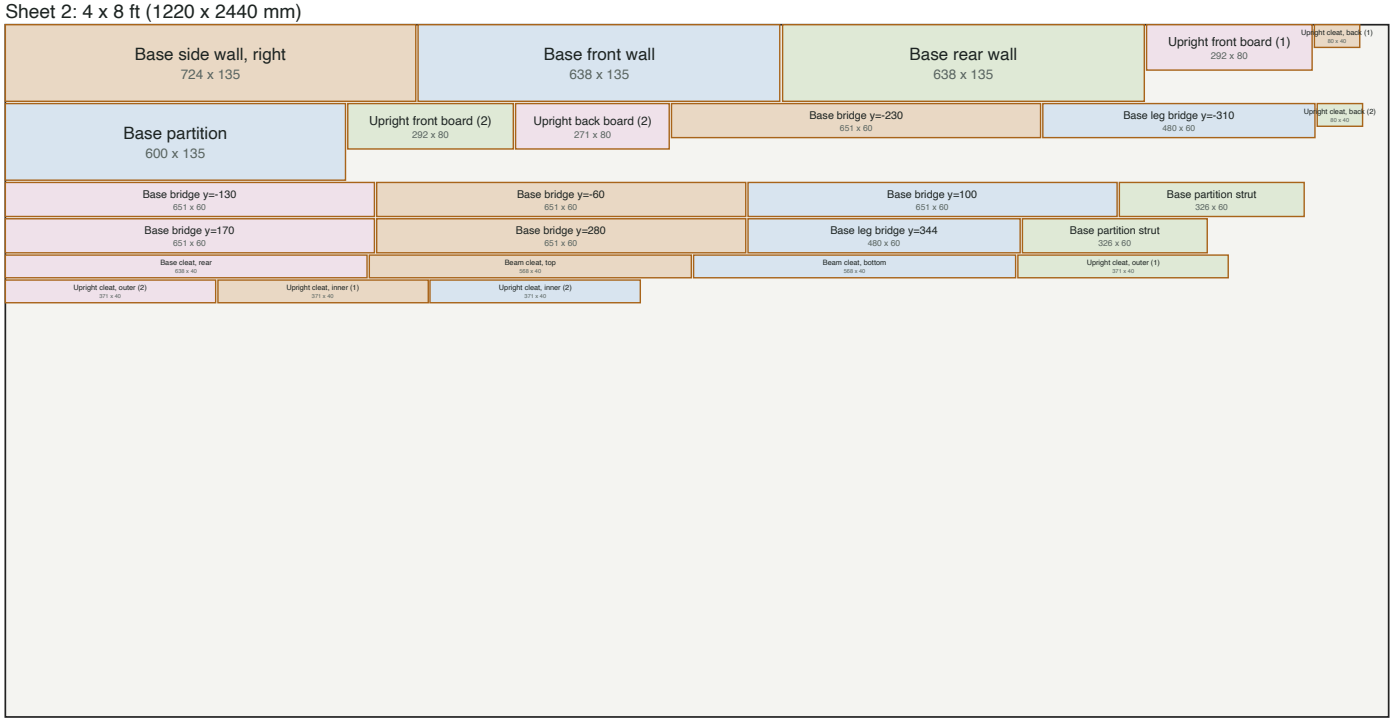
4. Cut the boards

Have the store rip the strips; crosscut at home. Write each board's name on its outside face, and mark the cast face (the melamine face the mix touches).

Sheet 1: 4 x 8 ft (1220 x 2440 mm)



Sheet 2: 4 x 8 ft (1220 x 2440 mm)



Board	Qty	Size (mm)	Notes
Base floor	1	718 x 843	both halves are cast on it
Base front wall	1	638 x 135	
Base rear wall	1	638 x 135	
Base side wall, left	1	724 x 135	
Base side wall, right	1	724 x 135	
Base partition	1	600 x 135	rear half only; comes out before the front half
Base partition strut	2	326 x 60	offcut, stands on edge; holds the partition foot (rear half only)
Base cleat, front	1	638 x 40	offcut strip, flat on the floor
Base cleat, rear	1	638 x 40	offcut strip, flat on the floor
Base cleat, left	1	724 x 40	offcut strip, flat on the floor
Base cleat, right	1	724 x 40	offcut strip, flat on the floor
Base bridge y=-230	1	651 x 60	
Base bridge y=-130	1	651 x 60	
Base bridge y=-60	1	651 x 60	
Base bridge y=100	1	651 x 60	
Base bridge y=170	1	651 x 60	
Base bridge y=280	1	651 x 60	
Base leg bridge y=-310	1	480 x 60	
Base leg bridge y=344	1	480 x 60	
Upright floor	2	198 x 541	
Upright side plate, outer	2	461 x 271 (profile)	cut to the profile
Upright side plate, inner	2	461 x 271 (profile)	cut to the profile
Upright back board	2	80 x 271	
Upright front board	2	80 x 292	ends cut square
Upright bridge	2	131 x 138	
Upright cleat, outer	2	371 x 40	offcut strip, flat on the floor
Upright cleat, inner	2	371 x 40	offcut strip, flat on the floor
Upright cleat, back	2	80 x 40	offcut strip, flat on the floor
Beam floor	1	648 x 264	
Beam top wall	1	568 x 165	
Beam bottom wall	1	568 x 165	
Beam end wall, left	1	146 x 165	
Beam end wall, right	1	146 x 165	
Beam cleat, top	1	568 x 40	offcut strip, flat on the floor
Beam cleat, bottom	1	568 x 40	offcut strip, flat on the floor

Board	Qty	Size (mm)	Notes
Beam cleat, left	1	146 x 40	offcut strip, flat on the floor
Beam cleat, right	1	146 x 40	offcut strip, flat on the floor

5. Mark and drill

1. Print the drilling templates (`templates/base-templates.pdf`, `templates/upright-templates.pdf`, `templates/beam-templates.pdf`) at **actual size** and check the 100 mm square.
2. Tape each board's pages onto it: printed edges on the board's edges, grid lines lined up.
3. Awl through every crosshair. Pencil the dashed **pour line** on the walls.
4. Drill from the marked face, at the size the color legend gives. Pilot the big ones; the 63.5 mm holes are a 2-1/2 in hole saw.

The appendix has every board as a drawing and a table, if you would rather measure.

Hole	Drill	For
3.2 mm	1/8 in	foot-gasket-groove, motor-relief, split-gasket-groove
3.4 mm	9/64 in	corner-L, corner-R, gutter-L, gutter-R
4.5 mm	11/64 in	back-edge, cleat, core post, corner bracket, corner-L, corner-R, draw-pocket-L, draw-pocket-R, gutter-L, gutter-R, leg holder, saddle, toe-filler, toe-plug, trough
5.5 mm	7/32 in	M5 insert screw, M5 screw into a rail bar, M5 screw into the face plate
6.5 mm	17/64 in	M6 screw into a pad, M6 screw into the face plate
9.6 mm	25/64 in	3/8 dowel, 3/8 dowel pin
10 mm	25/64 in	3/8-16 bolt into the top plate
13.1 mm	33/64 in	1/2 core rod, 1/2-13 bolt, base dowel hole
13.5 mm	17/32 in	column tie rod, from below

6. Cut the foam

Fine saw, hot wire or a long knife. Stack and glue two layers of 2 in foam for the 90 mm cores. Wrap each piece smooth in packing tape. Cut a snug 12 mm square hole 30 mm deep for each core's post and glue the post in.

Casting	Piece	Qty	Size and how
base-front	center-a	1	XPS block 100 x 73 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	center-b	1	XPS block 100 x 72 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.

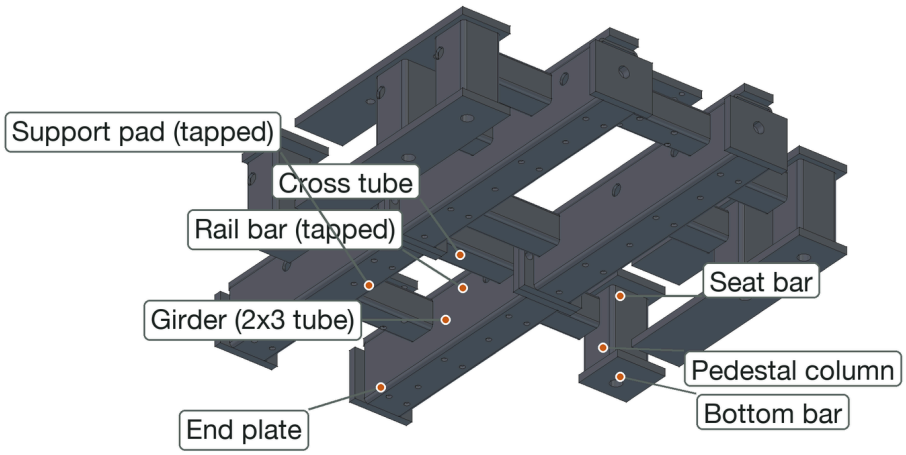
Casting	Piece	Qty	Size and how
base-rear	center	1	XPS block 88 x 190 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	outboard-L	1	XPS block 120 x 174 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	trough-band-L	1	XPS block 101 x 55 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	draw-rod-L	1	XPS block 20 x 255 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	foot-b-L	1	XPS block 39 x 99 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	knee-L	1	XPS block 50 x 140 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-a-L	1	XPS block 20 x 44 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-b-L	1	XPS block 20 x 60 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	rear-L	1	XPS block 85 x 52 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	outboard-R	1	XPS block 120 x 174 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	trough-band-R	1	XPS block 101 x 55 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	draw-rod-R	1	XPS block 20 x 255 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	foot-b-R	1	XPS block 39 x 99 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	knee-R	1	XPS block 50 x 140 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-a-R	1	XPS block 20 x 44 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-b-R	1	XPS block 20 x 60 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	rear-R	1	XPS block 85 x 52 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
## 7. Wax			

- 3-5 coats of paste wax on every face the mix touches, boards and removable printed pieces, then PVA. Don't wax pieces that stay in (core posts, cast-in parts): scuff and degrease them.
- Glue quarter-round into the floor-to-wall corners that make outside edges; silicone fillets elsewhere.
- Grease every screw, bolt and rod that comes out of the casting.

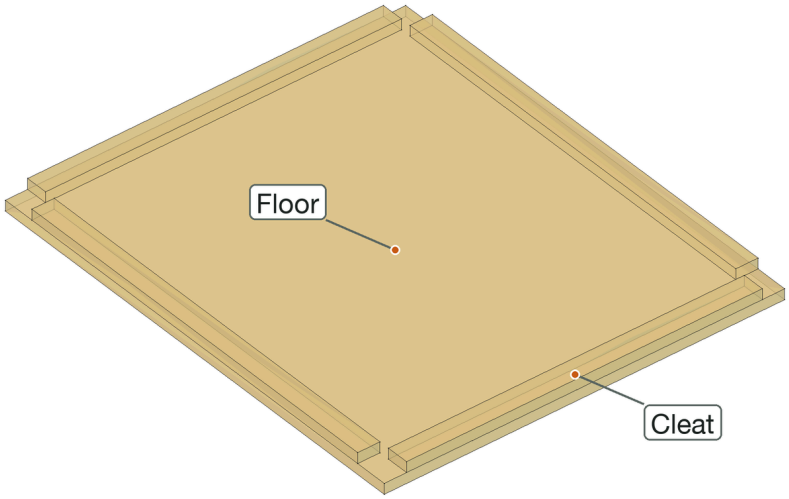
8. Assemble

Base, rear half (first)

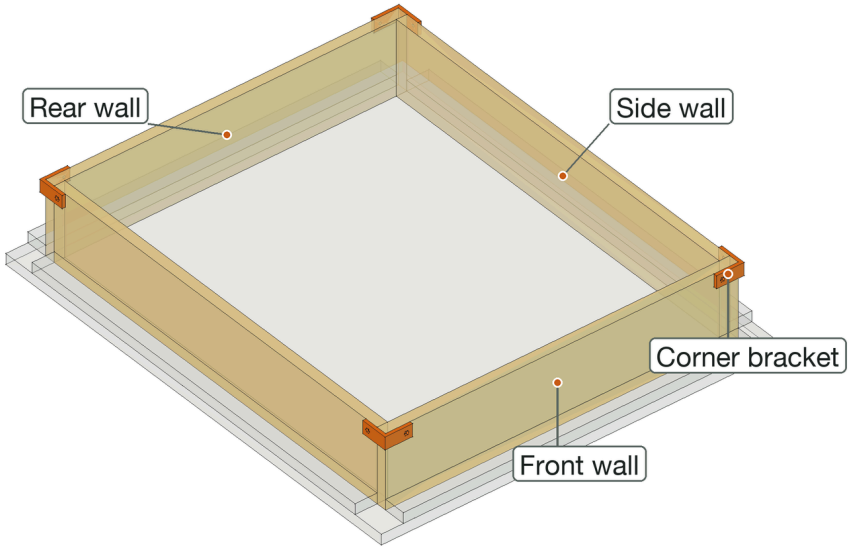
Cast upside down, top face on the floor; shown the right way up. New parts in color. Bolt each half's steel grid together first (the build guide, section 1), then it goes in as one piece.



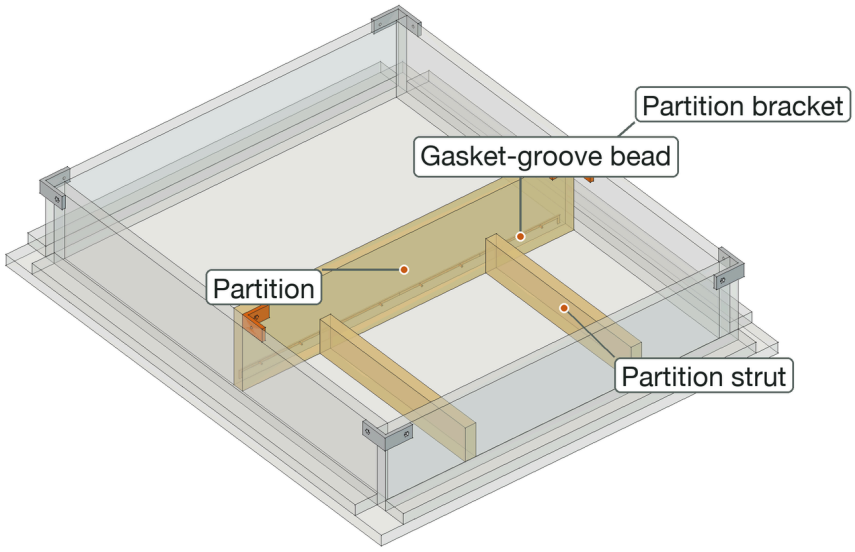
1. Floor on three battens. Bolt the cleats down.



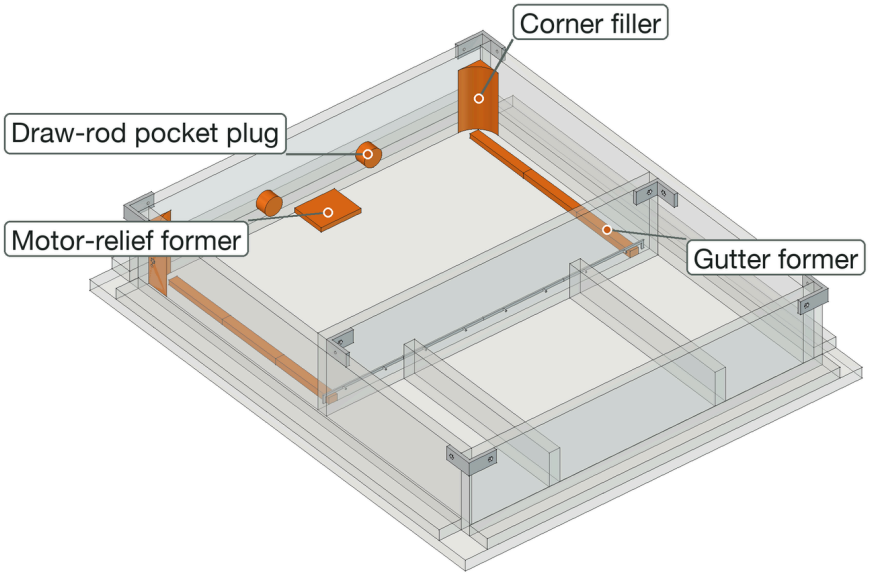
2. Walls against the cleats; corner brackets at the top. Diagonals equal within 2 mm, then tighten.



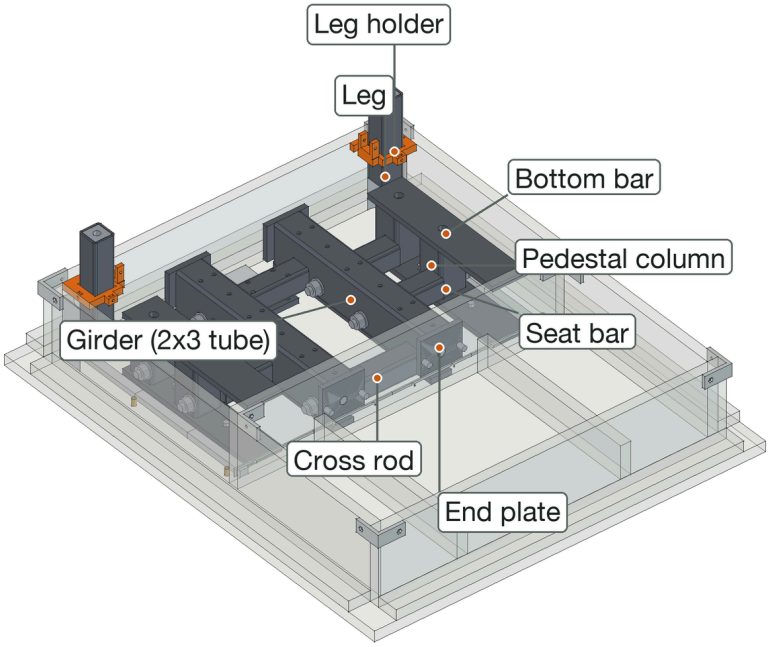
3. Partition between the side walls: two brackets, two struts to the front wall. Beads on its pins.



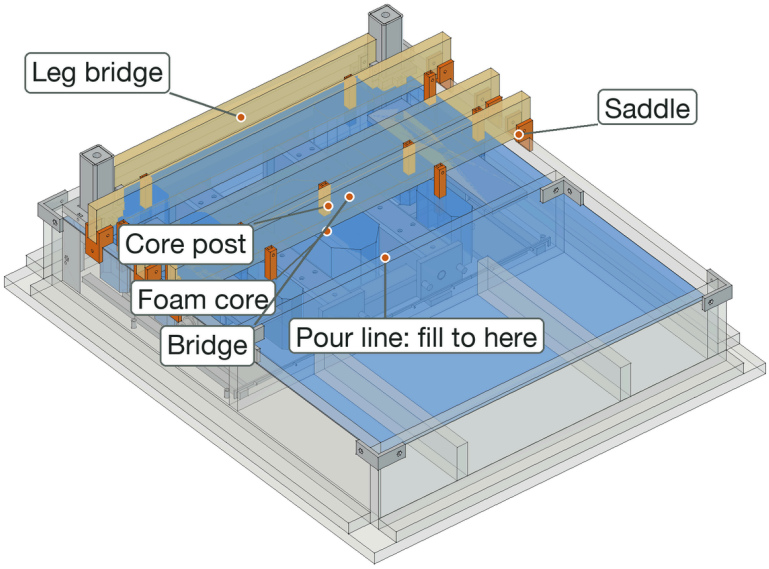
4. Gutter formers, motor relief, corner fillers and the draw-rod pocket plugs on.



5. The steel grid in, bolted up and upside down: its rail bars and the BK pad screwed to the floor (greased M5 and M6), its seats over the dowels, the split end plates against the partition with their dowel pins through it. Tape over the holes in the pedestals' bottom bars. Inserts; leg holders, legs set to height, pinch screws tight.

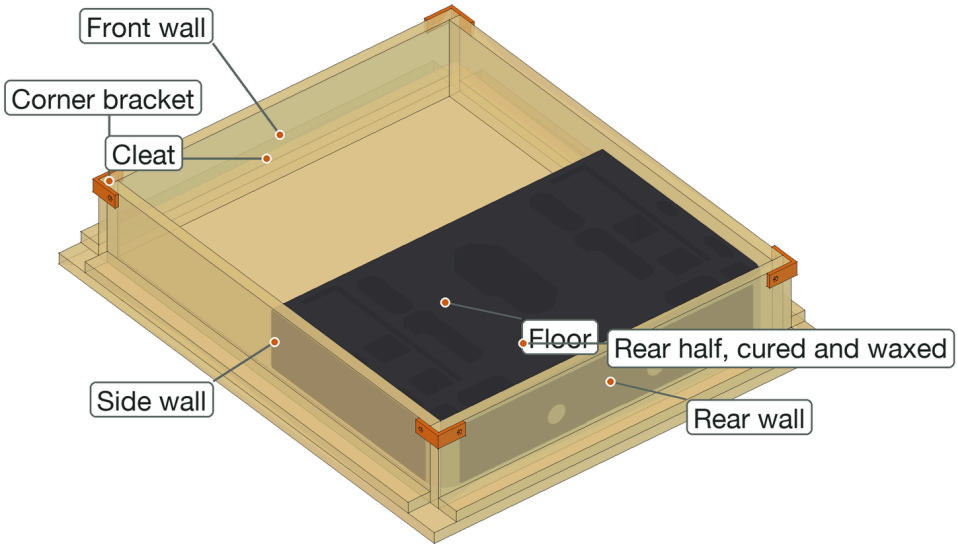


6. Pour the 20 mm skin. Foam cores on it between the steel, bridges into the saddles, each post bolted to its bridge. Fill to the pour line.

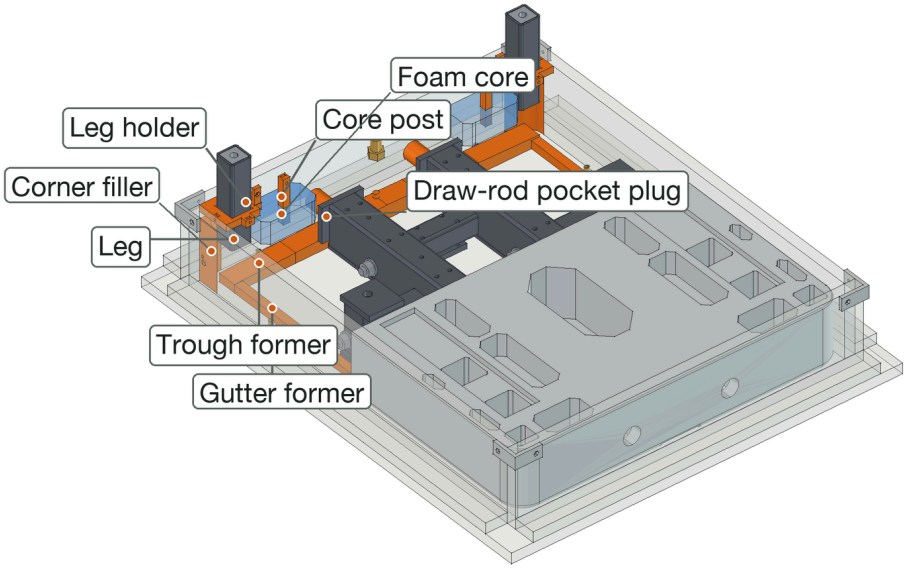


Base, front half (against the rear half)

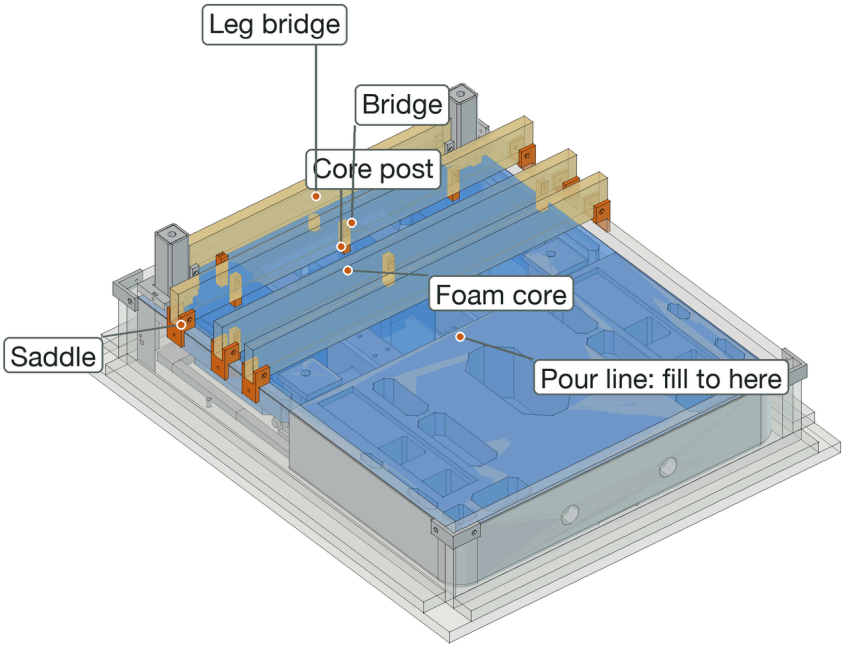
1. After the cure: bridges, partition and struts off. Pull the beads, tape the groove, and wax the rear half's split face, end plates and dowel pins.



2. Trough and gutter formers, corner fillers and pocket plugs on; the front grid in, its rail bars and BF pad screwed down and its split end plates onto the dowel pins; inserts and legs.

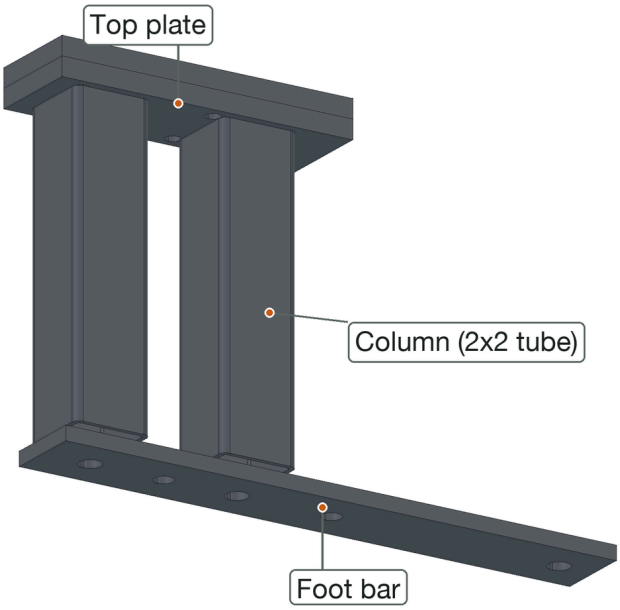


3. Skin, cores, bridges, fill. After the cure, take it apart and split the halves with wedges.

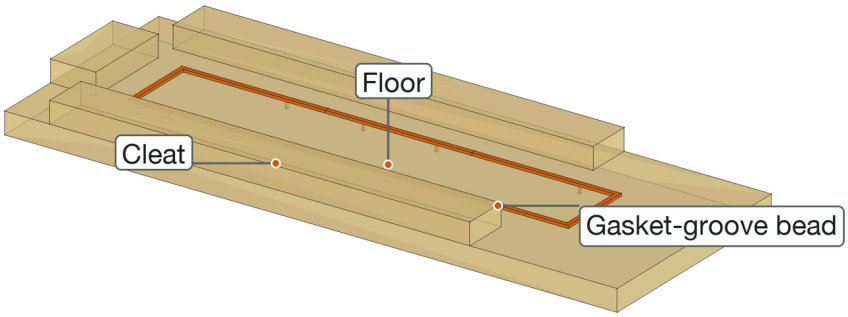


Uprights (two molds)

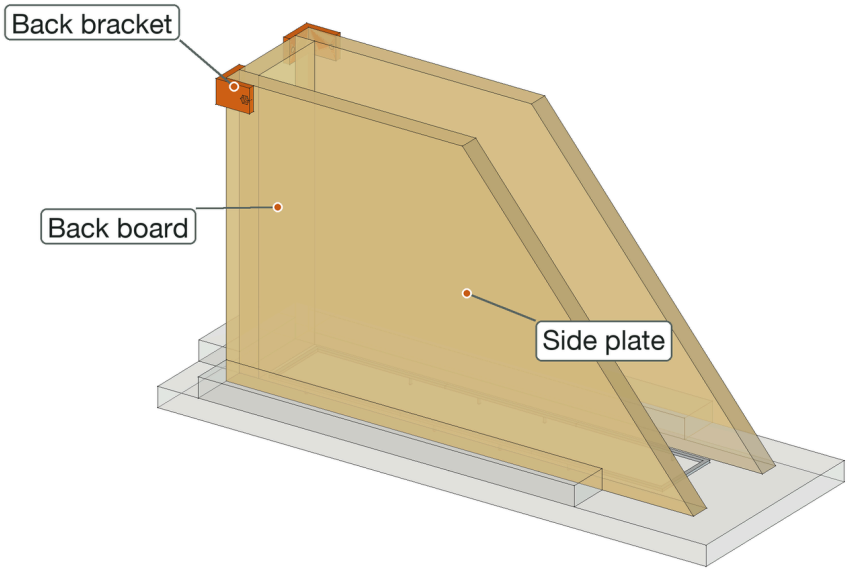
Cast foot down. Build two of everything. The skeleton: the foot bar, two columns and the top plate (two flat bars), clamped later by the column tie rods.



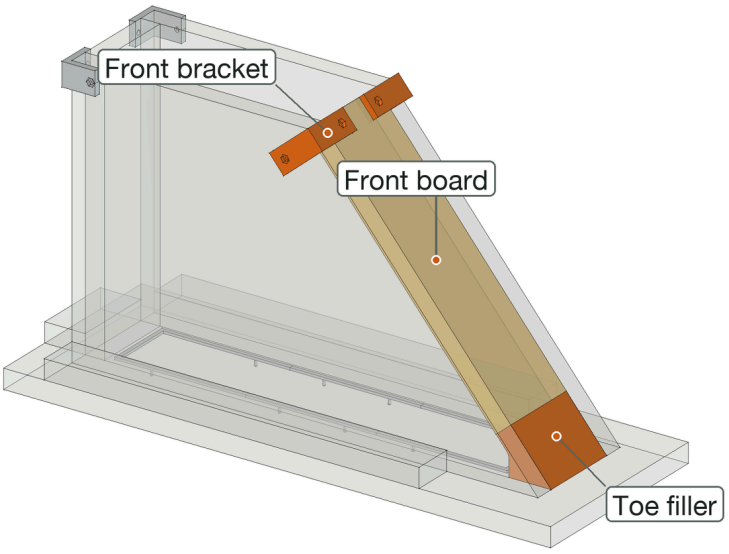
1. Gasket bead pinned on; cleats bolted down.



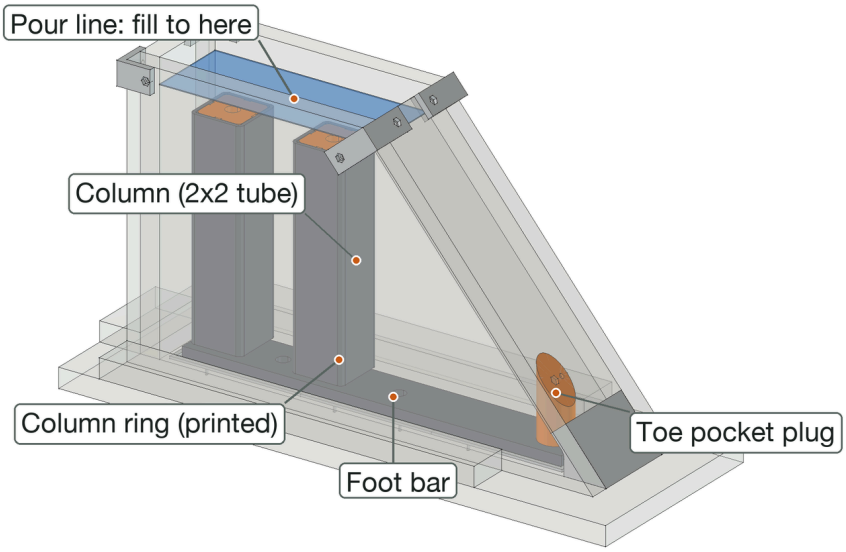
2. Side plates against their cleats, back board between them, two back brackets.



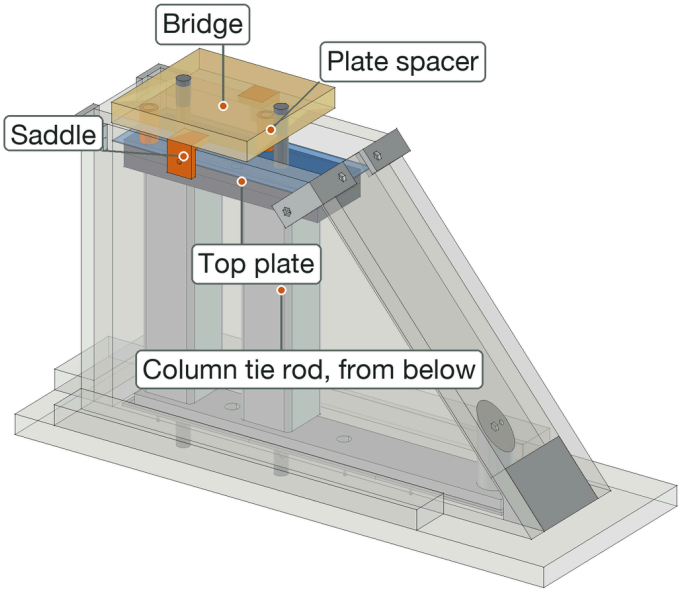
3. Toe filler bolted down, the front board's foot in its lip, two front brackets at the top.

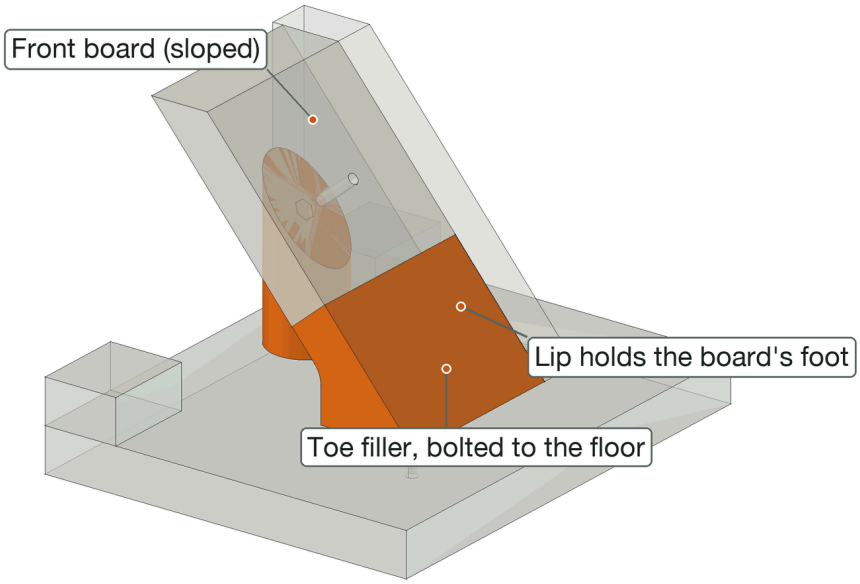


4. Mold on its four 2x4 blocks. The foot bar on the floor; the two columns on beds of thickened epoxy on it, the top plate and bridge set on them while it goes off, then lifted away; the toe pocket plug on the foot bar; the two dowel-hole bolts from underneath.



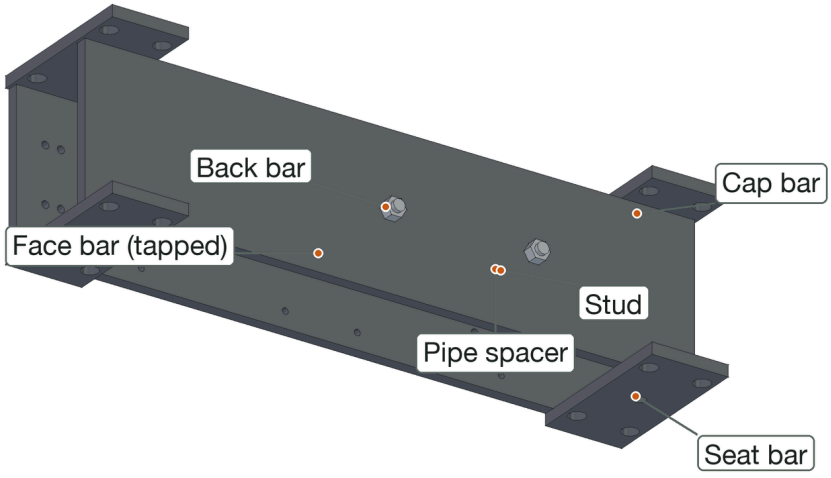
5. Fill to the line. Top plate (both bars) back on, hung from the bridge; bridge onto its saddles. Then the two column tie rods, greased, up from under the floor through the foot bar and the columns into the top plate: they center the columns while it cures.



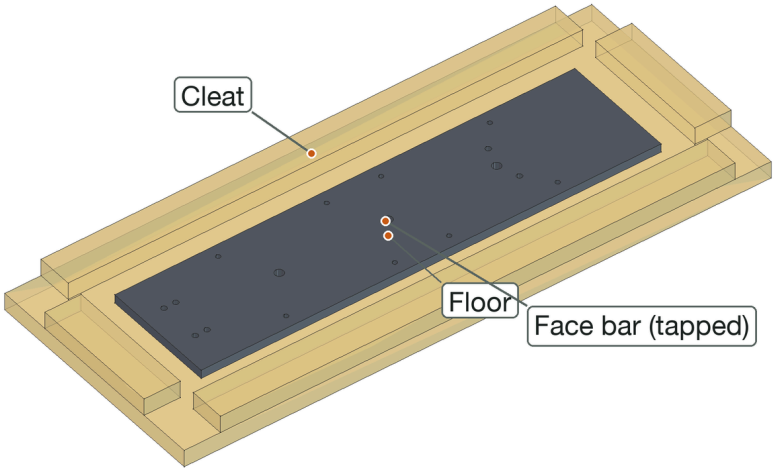


Beam

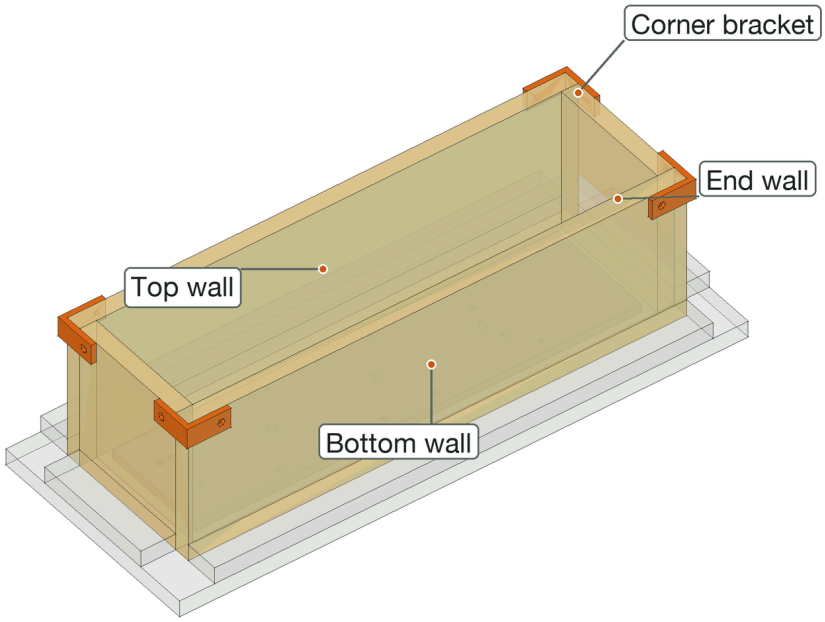
Cast front (rail) face down, back open. Bolt the steel together on the bench: the three studs into the face bar, a pipe spacer on each, the back bar over them and a nut on each.



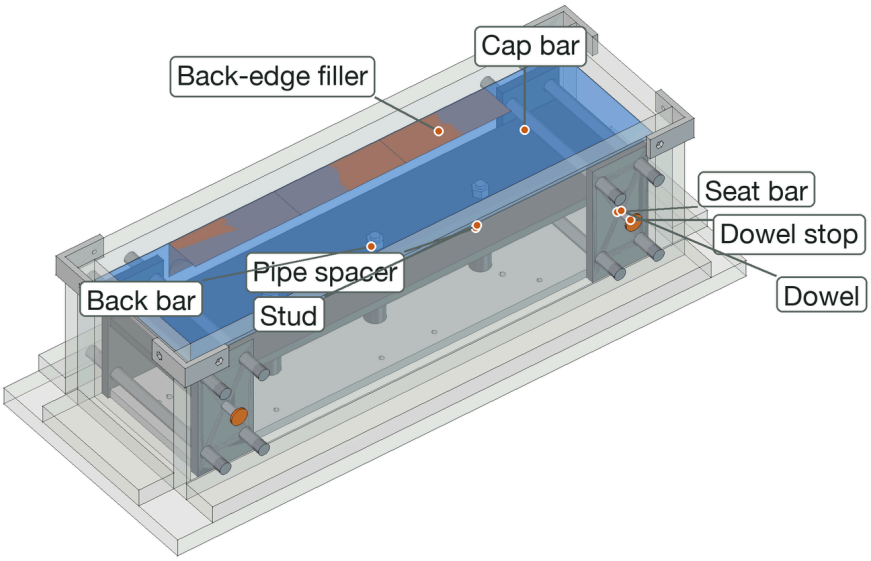
1. Cleats bolted down.



2. Walls against the cleats; four corner brackets.

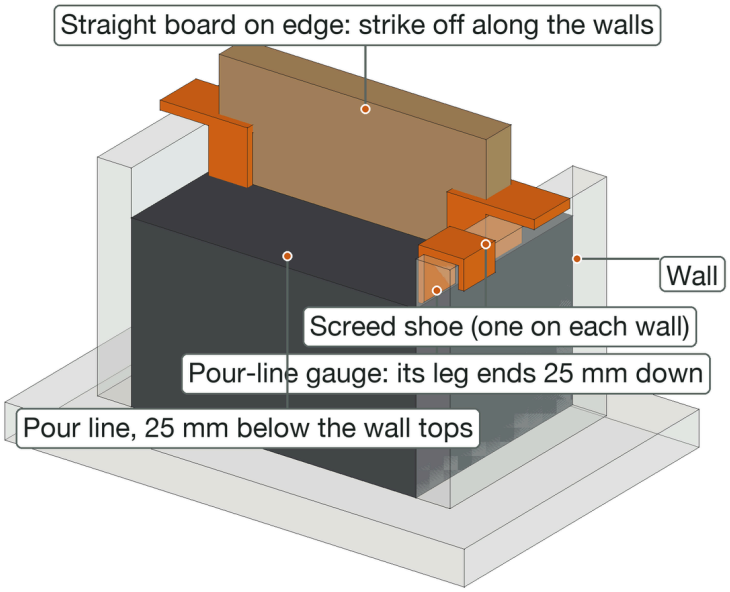


3. The steel in, face bar down: greased M5 and M6 screws up through the floor into its tapped holes; the back bar on its pipe spacers and studs; greased 1/2 in rods through the top wall, the cap bars, the seat bars and the bottom wall; back-edge filler on the top wall between the uprights; dowels on their stops. Fill to the line.



9. Pour

- Fill to the **pour line**, 25 mm below the wall tops (pencilled from the templates, or with the printed pour-line gauge clipped over a wall).
- Vibrate each layer until it stops bubbling; check the brackets halfway through.
- Strike the top off with a straight board in the two printed **screed shoes**: they hang it from the wall tops with its bottom edge on the pour line.



10. Take apart

1. After the full cure: bridges and saddles off, then corner brackets, then cleats; pull the walls off.
2. Screws, bolts and rods out from outside; lift the casting off the floor.
3. Unbolt and pull the formers and fillers.
4. Scrape and re-wax the boards; keep each mold's kit in a box. The kit is reused.

11. If something doesn't fit

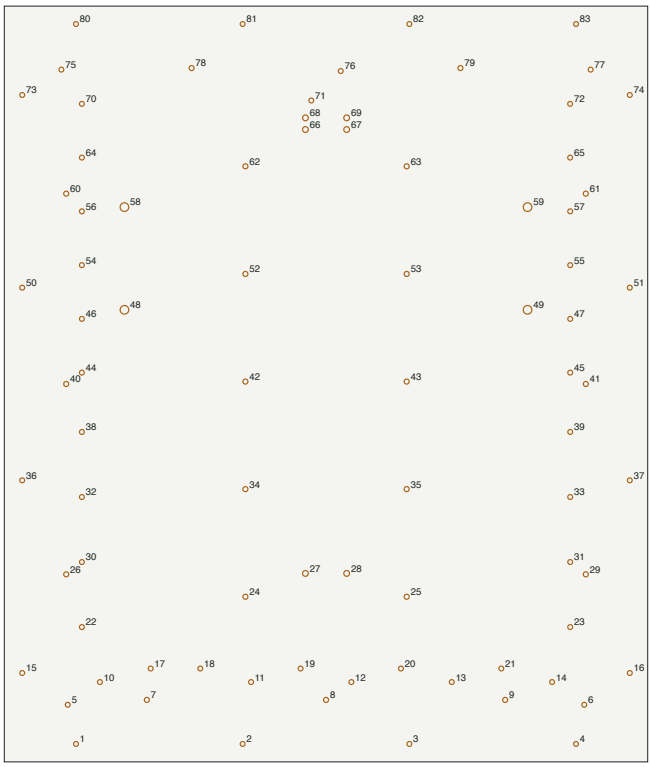
Problem	Fix
Nut won't slide under the hole	Clean the pocket with a nut on a bolt; warm the nut
Bracket or saddle loose on the board	Board thinner than set: tape shim, or set `mold_board` and reprint
Pin won't go in	Run the 1/8 in bit through again; trim the pin tip
Foam core moves	Tighten its post bolt; carpet tape under it
Printed part won't go on or into a steel tube	File the weld seam; sand the part. Several tight: caliper the tube, put its sizes in `frame_tube` / `post_tube`, re-run and reprint
Printed part loose on a tube	A wrap of tape round the part
Leak at a corner	Silicone fillet inside before waxing; check the brackets and diagonals
Former stuck after curing	Bolts out first, then a dowel and a light tap

Appendix: hole maps

Every board as seen from the face you mark: x from the left edge, y from the bottom edge (for a wall, the edge on the floor). The templates carry the same holes at full size.

Base

Base floor



0,0 · 718 x 843 mm · seen from the cast side

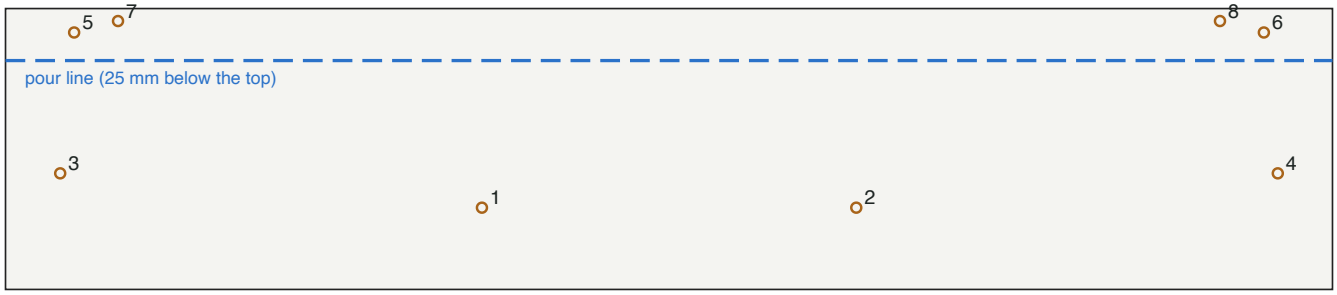
#	x	y	Hole	For
1	80	20	4.5	cleat
2	266	20	4.5	cleat
3	452	20	4.5	cleat
4	638	20	4.5	cleat
5	71	64	3.4	corner-R
6	647	64	3.4	corner-L
7	159	69	5.5	M5 insert screw
8	359	69	5.5	M5 insert screw
9	559	69	5.5	M5 insert screw
10	107	89	4.5	trough
11	275	89	4.5	trough
12	387	89	4.5	trough

#	x	y	Hole	For
13	499	89	4.5	trough
14	611	89	4.5	trough
15	20	99	4.5	cleat
16	698	99	4.5	cleat
17	163	104	4.5	trough
18	219	104	4.5	trough
19	331	104	4.5	trough
20	443	104	4.5	trough
21	555	104	4.5	trough
22	87	150	4.5	gutter-R
23	632	150	4.5	gutter-L
24	269	184	5.5	M5 screw into a rail bar
25	449	184	5.5	M5 screw into a rail bar
26	69	209	5.5	M5 insert screw
27	336	210	6.5	M6 screw into a pad
28	382	210	6.5	M6 screw into a pad
29	649	209	5.5	M5 insert screw
30	87	223	4.5	gutter-R
31	632	223	4.5	gutter-L
32	87	295	3.4	gutter-R
33	632	295	3.4	gutter-L
34	269	304	5.5	M5 screw into a rail bar
35	449	304	5.5	M5 screw into a rail bar
36	20	314	4.5	cleat
37	698	314	4.5	cleat
38	87	368	3.4	gutter-R
39	632	368	3.4	gutter-L
40	69	421	5.5	M5 insert screw
41	649	421	5.5	M5 insert screw
42	269	424	5.5	M5 screw into a rail bar
43	449	424	5.5	M5 screw into a rail bar
44	87	434	3.4	gutter-R
45	632	434	3.4	gutter-L
46	87	494	3.4	gutter-R
47	632	494	3.4	gutter-L

#	x	y	Hole	For
48	134	504	9.6	3/8 dowel
49	584	504	9.6	3/8 dowel
50	20	529	4.5	cleat
51	698	529	4.5	cleat
52	269	544	5.5	M5 screw into a rail bar
53	449	544	5.5	M5 screw into a rail bar
54	87	554	3.4	gutter-R
55	632	554	3.4	gutter-L
56	87	614	3.4	gutter-R
57	632	614	3.4	gutter-L
58	134	619	9.6	3/8 dowel
59	584	619	9.6	3/8 dowel
60	69	634	5.5	M5 insert screw
61	649	634	5.5	M5 insert screw
62	269	664	5.5	M5 screw into a rail bar
63	449	664	5.5	M5 screw into a rail bar
64	87	674	3.4	gutter-R
65	632	674	3.4	gutter-L
66	336	705	6.5	M6 screw into a pad
67	382	705	6.5	M6 screw into a pad
68	336	718	6.5	M6 screw into a pad
69	382	718	6.5	M6 screw into a pad
70	87	734	3.4	gutter-R
71	343	737	3.2	motor-relief
72	632	734	3.4	gutter-L
73	20	744	4.5	cleat
74	698	744	4.5	cleat
75	64	772	3.4	corner-R
76	375	770	3.2	motor-relief
77	654	772	3.4	corner-L
78	209	774	5.5	M5 insert screw
79	509	774	5.5	M5 insert screw
80	80	823	4.5	cleat
81	266	823	4.5	cleat
82	452	823	4.5	cleat

#	x	y	Hole	For
83	638	823	4.5	cleat

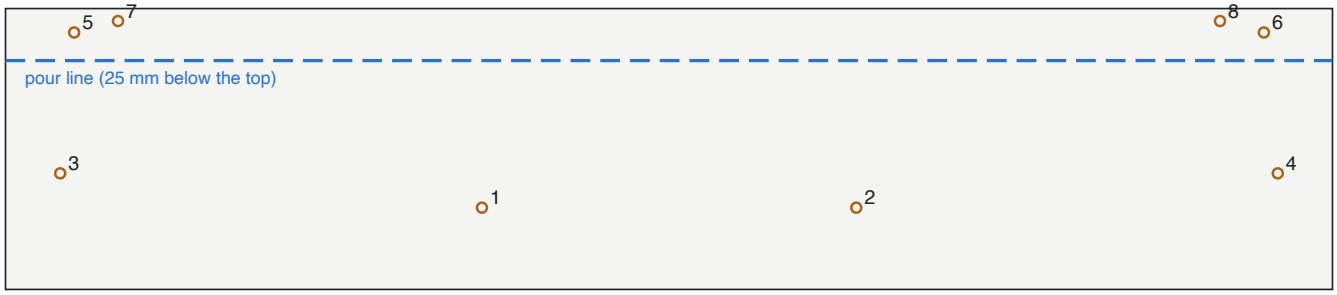
Base front wall



0,0 · 638 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	229	39	4.5	draw-pocket-L
2	409	39	4.5	draw-pocket-R
3	26	56	4.5	corner-L
4	612	56	4.5	corner-R
5	33	124	4.5	corner bracket
6	605	124	4.5	corner bracket
7	54	129	4.5	leg holder
8	584	129	4.5	leg holder

Base rear wall

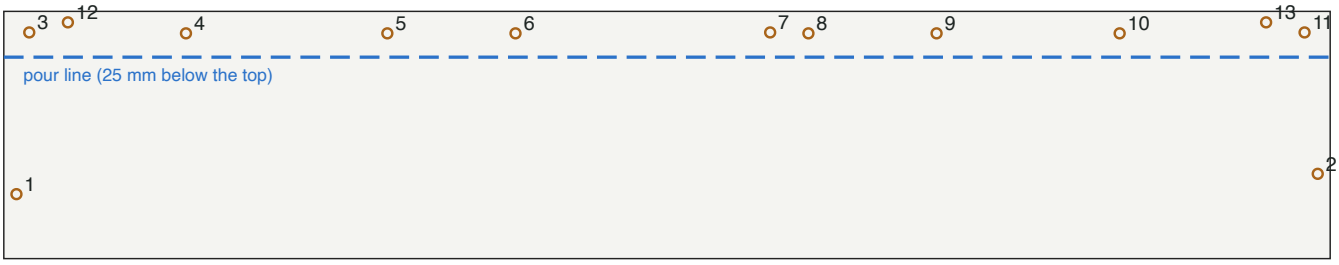


0,0 · 638 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	229	39	4.5	draw-pocket-R
2	409	39	4.5	draw-pocket-L
3	26	56	4.5	corner-R
4	612	56	4.5	corner-L
5	33	124	4.5	corner bracket

#	x	y	Hole	For
6	605	124	4.5	corner bracket
7	54	129	4.5	leg holder
8	584	129	4.5	leg holder

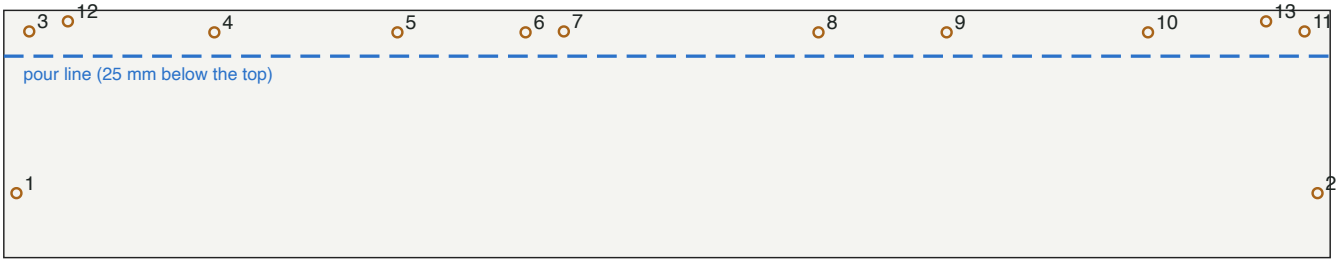
Base side wall, left



0,0 · 724 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	7	35	4.5	corner-L
2	718	46	4.5	corner-L
3	14	124	4.5	corner bracket
4	100	123	4.5	saddle
5	210	123	4.5	saddle
6	280	123	4.5	saddle
7	419	124	4.5	corner bracket
8	440	123	4.5	saddle
9	510	123	4.5	saddle
10	610	123	4.5	saddle
11	711	124	4.5	corner bracket
12	35	129	4.5	leg holder
13	690	129	4.5	leg holder

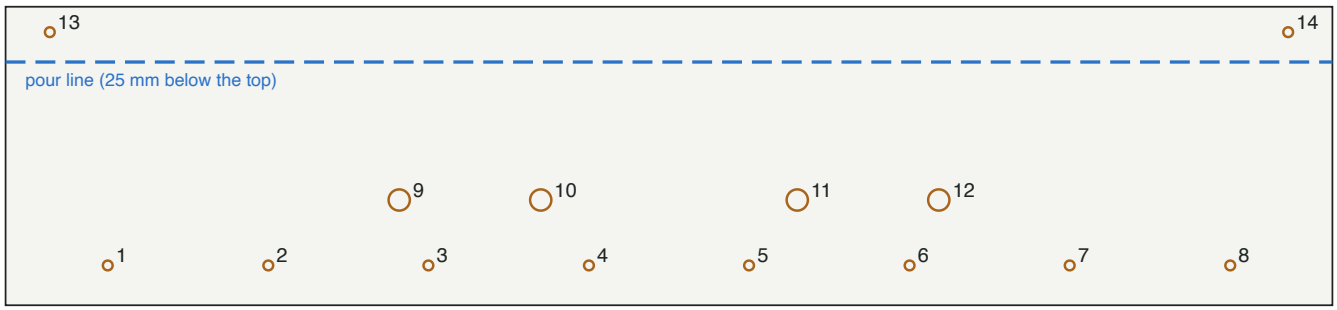
Base side wall, right



0,0 · 724 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	7	35	4.5	corner-R
2	718	35	4.5	corner-R
3	14	124	4.5	corner bracket
4	115	123	4.5	saddle
5	215	123	4.5	saddle
6	285	123	4.5	saddle
7	306	124	4.5	corner bracket
8	445	123	4.5	saddle
9	515	123	4.5	saddle
10	625	123	4.5	saddle
11	711	124	4.5	corner bracket
12	35	129	4.5	leg holder
13	690	129	4.5	leg holder

Base partition



0,0 · 600 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	46	18	3.2	split-gasket-groove
2	119	18	3.2	split-gasket-groove
3	191	18	3.2	split-gasket-groove
4	264	18	3.2	split-gasket-groove
5	336	18	3.2	split-gasket-groove
6	409	18	3.2	split-gasket-groove
7	481	18	3.2	split-gasket-groove
8	554	18	3.2	split-gasket-groove
9	178	48	9.6	3/8 dowel pin (split)
10	242	48	9.6	3/8 dowel pin (split)
11	358	48	9.6	3/8 dowel pin (split)

#	x	y	Hole	For
12	422	48	9.6	3/8 dowel pin (split)
13	20	124	4.5	corner bracket
14	580	124	4.5	corner bracket

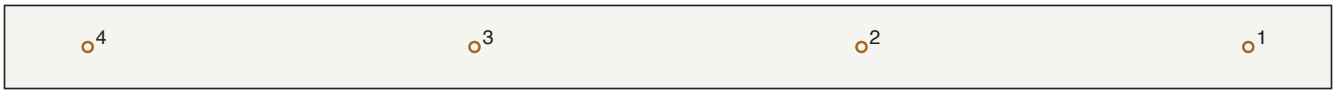
Base cleat, front



0,0 · 638 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	598	20	4.5	cleat
2	412	20	4.5	cleat
3	226	20	4.5	cleat
4	40	20	4.5	cleat

Base cleat, rear



0,0 · 638 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	598	20	4.5	cleat
2	412	20	4.5	cleat
3	226	20	4.5	cleat
4	40	20	4.5	cleat

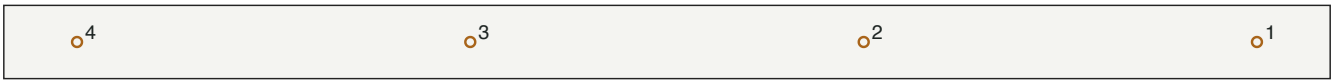
Base cleat, left



0,0 · 724 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	684	20	4.5	cleat
2	470	20	4.5	cleat
3	255	20	4.5	cleat
4	40	20	4.5	cleat

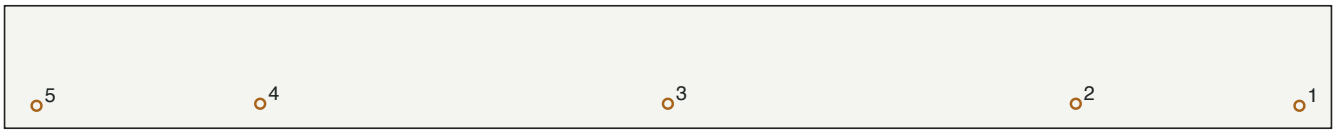
Base cleat, right



0,0 · 724 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	684	20	4.5	cleat
2	470	20	4.5	cleat
3	255	20	4.5	cleat
4	40	20	4.5	cleat

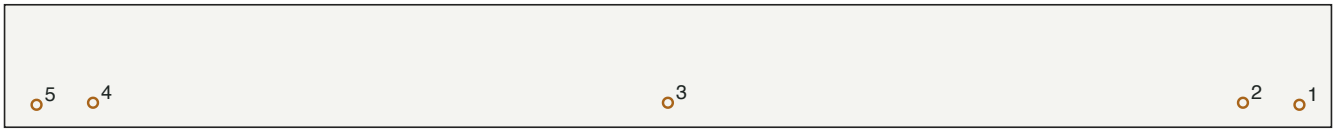
Base bridge y=-230



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	525	12	4.5	core post
3	325	12	4.5	core post
4	125	12	4.5	core post
5	16	11	4.5	saddle

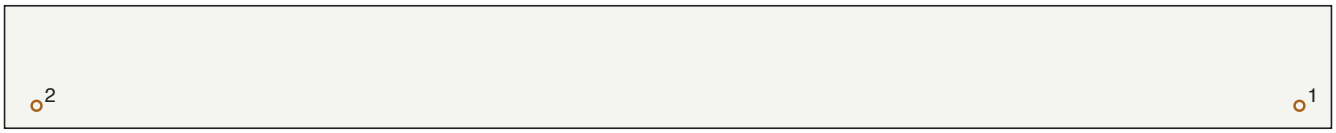
Base bridge y=-130



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	608	12	4.5	core post
3	325	12	4.5	core post
4	43	12	4.5	core post
5	16	11	4.5	saddle

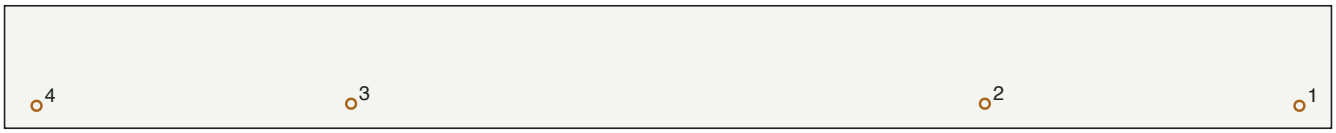
Base bridge y=-60



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	16	11	4.5	saddle

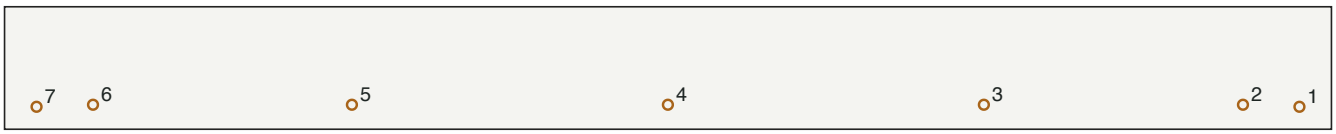
Base bridge y=100



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	481	12	4.5	core post
3	170	12	4.5	core post
4	16	11	4.5	saddle

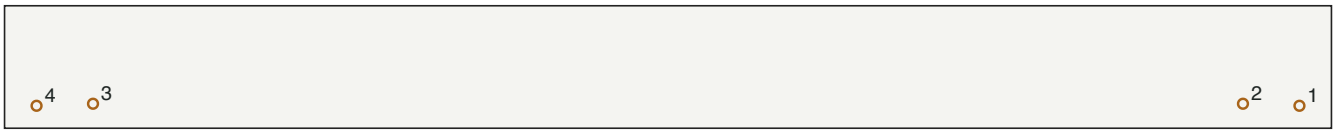
Base bridge y=170



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	608	12	4.5	core post
3	480	12	4.5	core post
4	325	12	4.5	core post
5	170	12	4.5	core post
6	43	12	4.5	core post
7	16	11	4.5	saddle

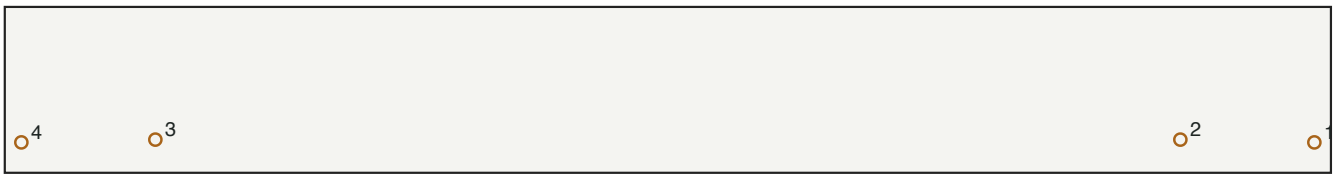
Base bridge y=280



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	608	12	4.5	core post
3	43	12	4.5	core post
4	16	11	4.5	saddle

Base leg bridge y=-310



0,0 · 480 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	474	11	4.5	leg holder
2	426	12	4.5	core post
3	54	12	4.5	core post
4	6	11	4.5	leg holder

Base leg bridge y=344

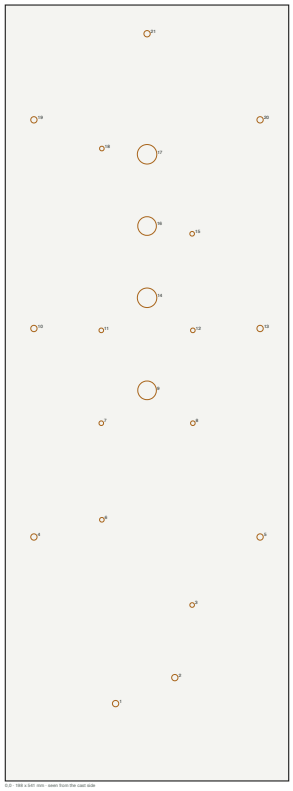


0,0 · 480 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	474	11	4.5	leg holder
2	428	12	4.5	core post
3	52	12	4.5	core post
4	6	11	4.5	leg holder

Uprights

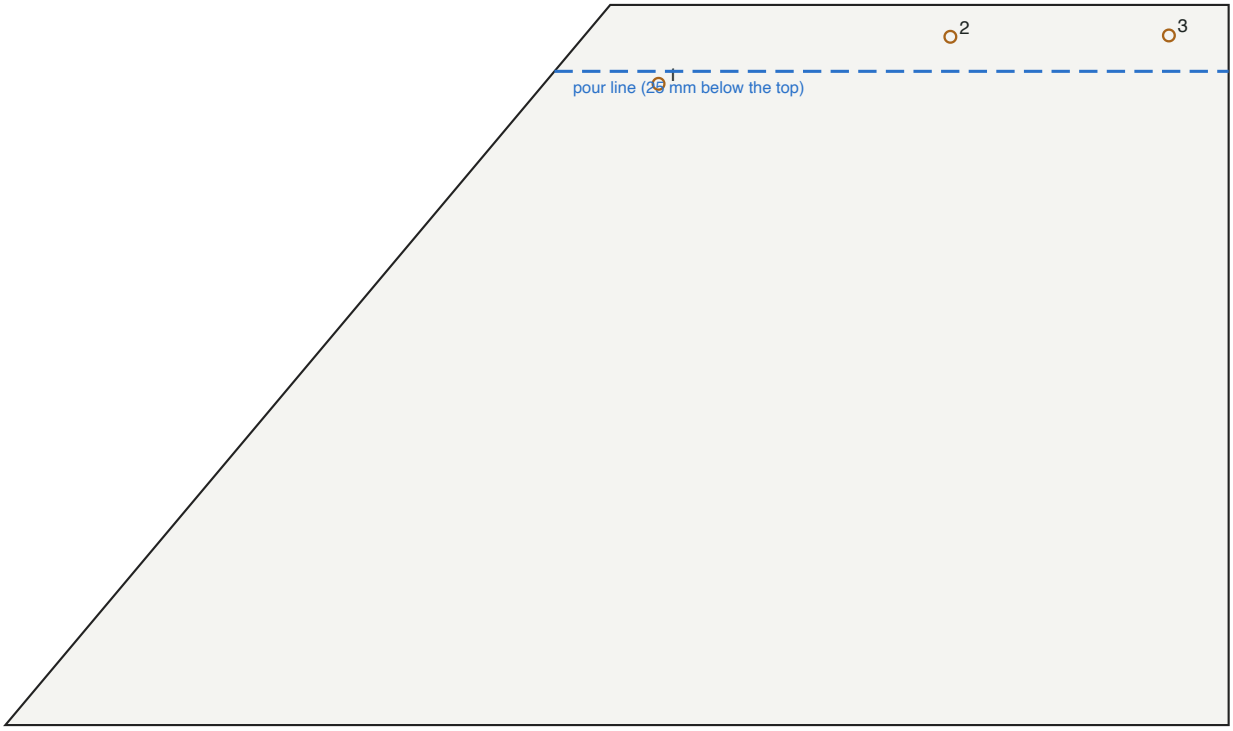
Upright floor (x2)



#	x	y	Hole	For
1	77	54	4.5	toe-filler
2	119	72	4.5	toe-filler
3	131	123	3.2	foot-gasket-groove
4	20	170	4.5	cleat
5	178	170	4.5	cleat
6	67	182	3.2	foot-gasket-groove
7	67	249	3.2	foot-gasket-groove
8	131	249	3.2	foot-gasket-groove
9	99	272	13.1	1/2-13 bolt, base dowel hole
10	20	315	4.5	cleat
11	67	314	3.2	foot-gasket-groove
12	131	314	3.2	foot-gasket-groove
13	178	315	4.5	cleat
14	99	337	13.5	column tie rod, from below
15	131	381	3.2	foot-gasket-groove
16	99	387	13.1	1/2-13 bolt, base dowel hole

#	x	y	Hole	For
17	99	437	13.5	column tie rod, from below
18	67	441	3.2	foot-gasket-groove
19	20	461	4.5	cleat
20	178	461	4.5	cleat
21	99	521	4.5	cleat

Upright side plate, outer (x2)

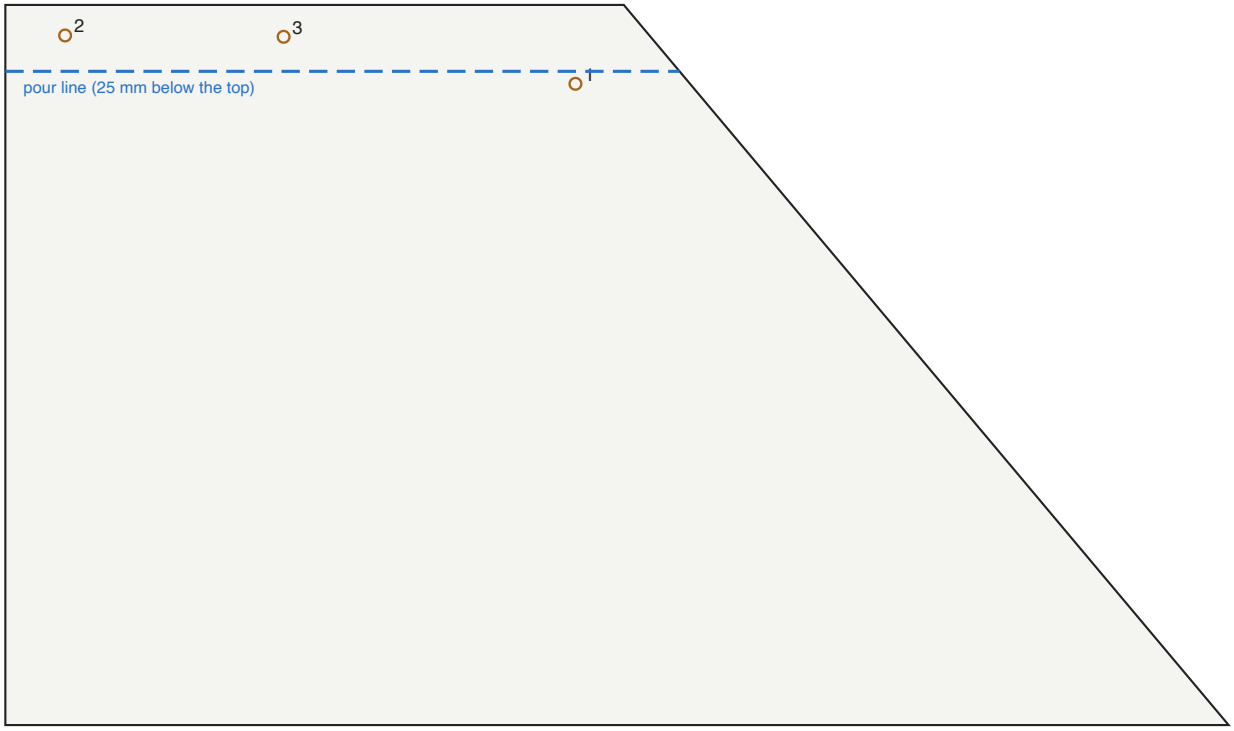


0,0 · 461 x 271 mm · seen from the cast side

#	x	y	Hole	For
1	246	241	4.5	corner bracket
2	356	259	4.5	saddle
3	438	260	4.5	corner bracket

Outline corners (x, y): (0, 0), (461, 0), (461, 271), (228, 271)

Upright side plate, inner (x2)

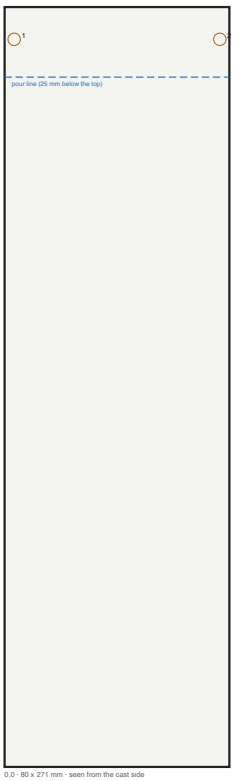


0,0 · 461 x 271 mm · seen from the cast side

#	x	y	Hole	For
1	215	241	4.5	corner bracket
2	22	260	4.5	corner bracket
3	105	259	4.5	saddle

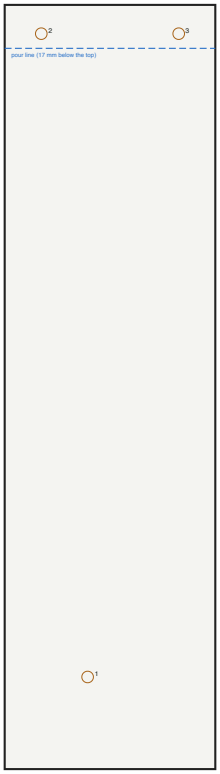
Outline corners (x, y): (461, 0), (0, 0), (0, 271), (233, 271)

Upright back board (x2)



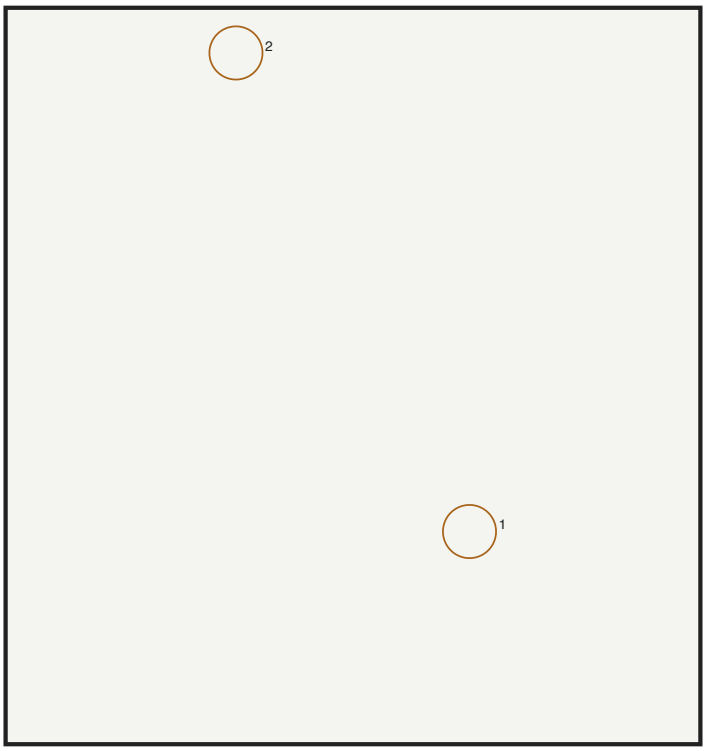
#	x	y	Hole	For
1	3	260	4.5	corner bracket
2	77	260	4.5	corner bracket

Upright front board (x2)



#	x	y	Hole	For
1	32	35	4.5	toe-plug
2	14	281	4.5	corner bracket
3	66	281	4.5	corner bracket

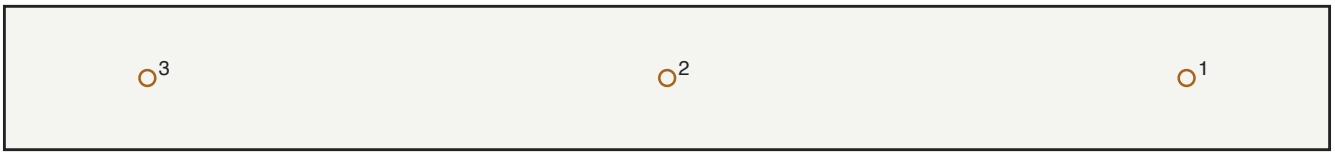
Upright bridge (x2)



0,0 · 131 x 138 mm · seen from the outside (top)

#	x	y	Hole	For
1	87	40	10	3/8-16 bolt into the top plate
2	43	130	10	3/8-16 bolt into the top plate

Upright cleat, outer (x2)



0,0 · 371 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	331	20	4.5	cleat
2	185	20	4.5	cleat
3	40	20	4.5	cleat

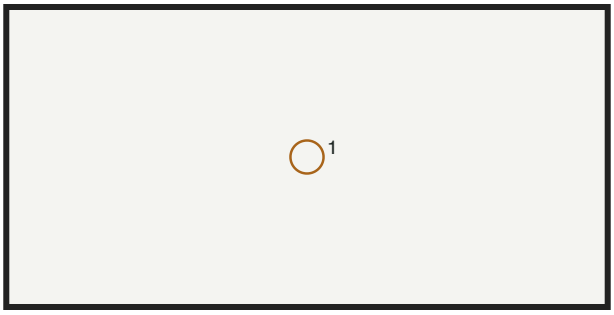
Upright cleat, inner (x2)



0,0 · 371 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	331	20	4.5	cleat
2	185	20	4.5	cleat
3	40	20	4.5	cleat

Upright cleat, back (x2)

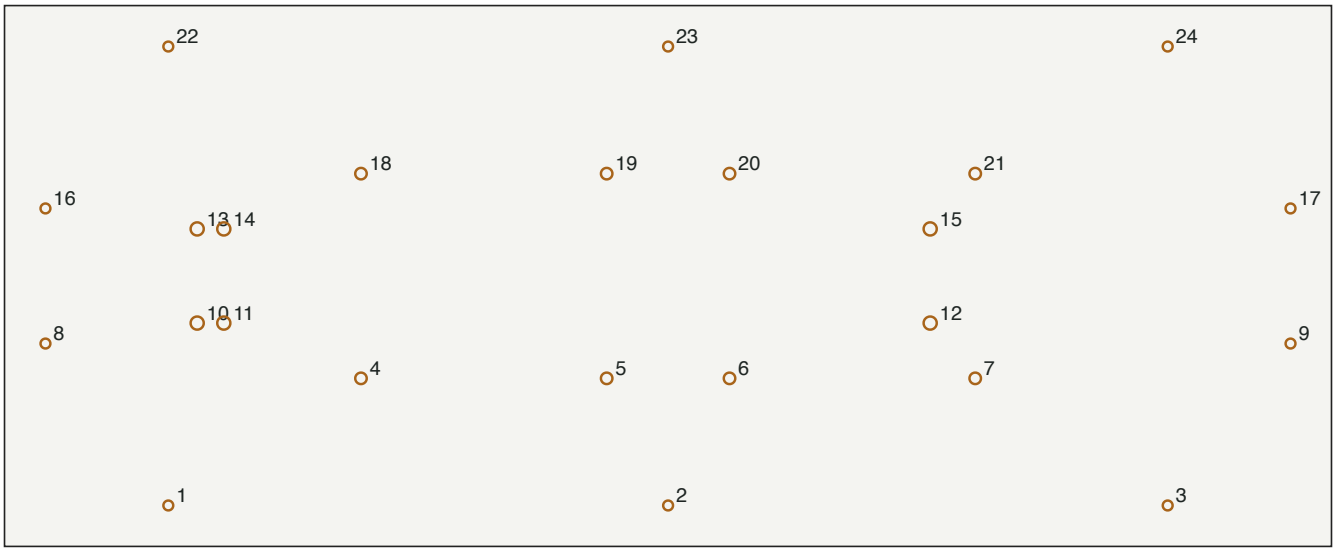


0,0 · 80 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	40	20	4.5	cleat

Beam

Beam floor

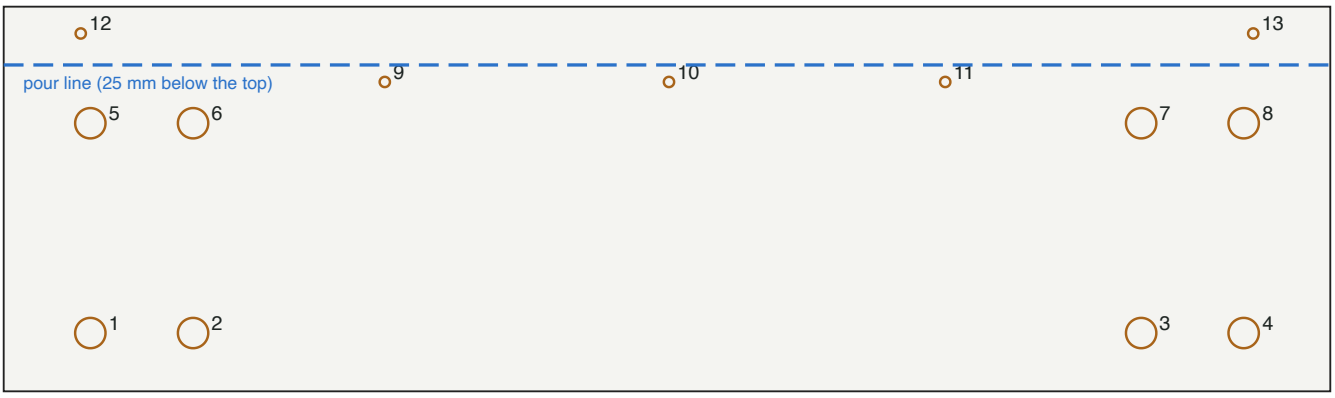


0,0 · 648 x 264 mm · seen from the cast side

#	x	y	Hole	For
1	80	20	4.5	cleat
2	324	20	4.5	cleat
3	568	20	4.5	cleat
4	174	82	5.5	M5 screw into the face plate
5	294	82	5.5	M5 screw into the face plate
6	354	82	5.5	M5 screw into the face plate
7	474	82	5.5	M5 screw into the face plate
8	20	99	4.5	cleat
9	628	99	4.5	cleat
10	94	109	6.5	M6 screw into the face plate
11	107	109	6.5	M6 screw into the face plate
12	452	109	6.5	M6 screw into the face plate
13	94	155	6.5	M6 screw into the face plate
14	107	155	6.5	M6 screw into the face plate
15	452	155	6.5	M6 screw into the face plate
16	20	165	4.5	cleat
17	628	165	4.5	cleat
18	174	182	5.5	M5 screw into the face plate
19	294	182	5.5	M5 screw into the face plate
20	354	182	5.5	M5 screw into the face plate
21	474	182	5.5	M5 screw into the face plate
22	80	244	4.5	cleat
23	324	244	4.5	cleat

#	x	y	Hole	For
24	568	244	4.5	cleat

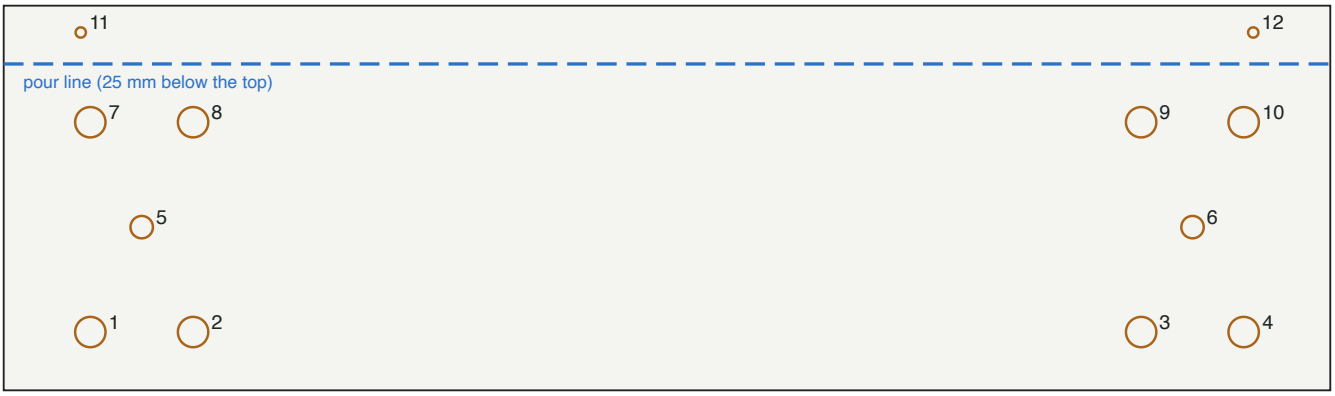
Beam top wall



0,0 · 568 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	37	25	13.1	1/2 core rod
2	81	25	13.1	1/2 core rod
3	487	25	13.1	1/2 core rod
4	531	25	13.1	1/2 core rod
5	37	115	13.1	1/2 core rod
6	81	115	13.1	1/2 core rod
7	487	115	13.1	1/2 core rod
8	531	115	13.1	1/2 core rod
9	163	133	4.5	back-edge
10	285	133	4.5	back-edge
11	403	133	4.5	back-edge
12	33	154	4.5	corner bracket
13	535	154	4.5	corner bracket

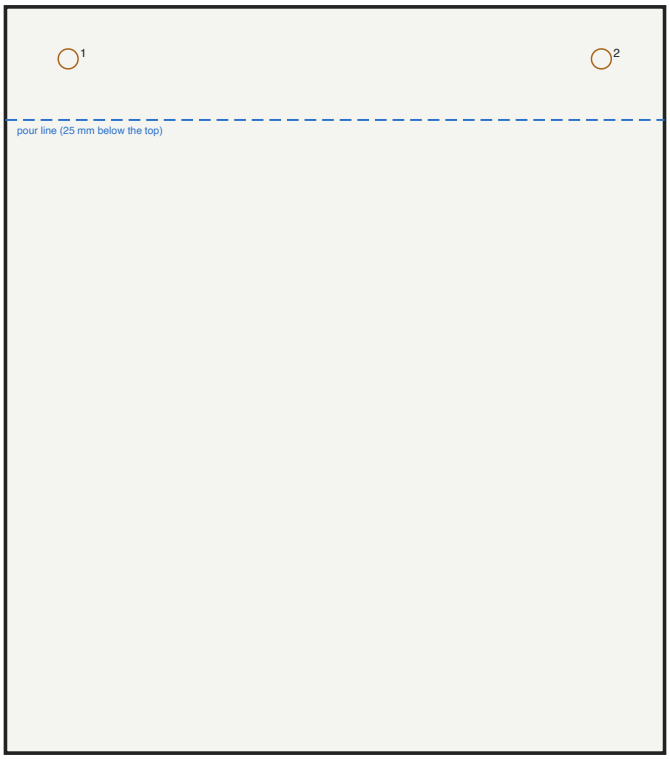
Beam bottom wall



0,0 · 568 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	37	25	13.1	1/2 core rod
2	81	25	13.1	1/2 core rod
3	487	25	13.1	1/2 core rod
4	531	25	13.1	1/2 core rod
5	59	70	9.6	3/8 dowel
6	509	70	9.6	3/8 dowel
7	37	115	13.1	1/2 core rod
8	81	115	13.1	1/2 core rod
9	487	115	13.1	1/2 core rod
10	531	115	13.1	1/2 core rod
11	33	154	4.5	corner bracket
12	535	154	4.5	corner bracket

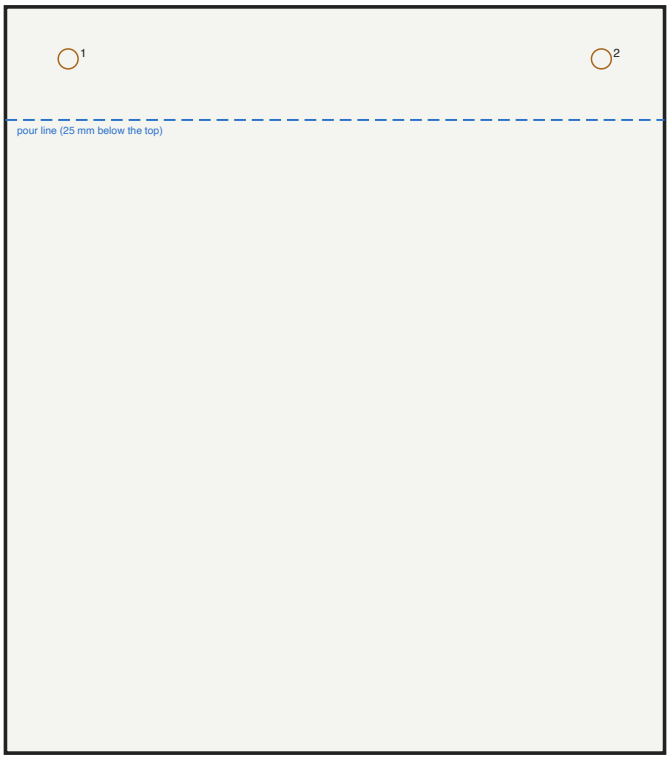
Beam end wall, left



0,0 · 146 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	14	154	4.5	corner bracket
2	132	154	4.5	corner bracket

Beam end wall, right



0,0 · 146 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	14	154	4.5	corner bracket
2	132	154	4.5	corner bracket

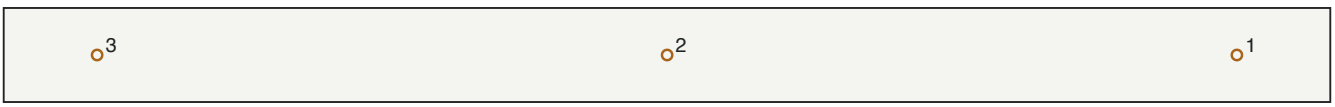
Beam cleat, top



0,0 · 568 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	528	20	4.5	cleat
2	284	20	4.5	cleat
3	40	20	4.5	cleat

Beam cleat, bottom



0,0 · 568 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	528	20	4.5	cleat
2	284	20	4.5	cleat
3	40	20	4.5	cleat

Beam cleat, left



0,0 · 146 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	106	20	4.5	cleat
2	40	20	4.5	cleat

Beam cleat, right



0,0 · 146 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	106	20	4.5	cleat
2	40	20	4.5	cleat